



Modeling the coupling of China's multi-timescale electricity markets during the transition towards decarbonization and marketization

Peng Wang^a, Wentao Wang^a, Kai Jiang^{a,*}, Yixin Cheng^a, Tengxi Zhang^b, Xuesong Li^c

^a State Key Laboratory of Alternate Electrical Power System with Renewable Energy Sources, North China Electric Power University, Beijing, China

^b School of Control and Computer Engineering, North China Electric Power University, Beijing, China

^c State Grid Jiangsu Electric Power Co., Ltd, Nanjing, China

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ABSTRACT

During the transition of the power system towards marketization, the delivery method of medium- and long-term (MLT) contracts in the spot electricity market fluctuates between physical delivery and financial delivery. It is challenging to optimize the coupling form for MLT contracts and the spot electricity market based on China's development stage. To tackle this issue, a bi-level model for the electricity-carbon market is proposed. The upper-level model captures the strategic decisions of multiple generation companies (GenCos), while the lower-level model focuses on market clearing. The delivery method of MLT contracts is categorized as physical, transitional, and financial and is incorporated into the model. To evaluate the model simulation, an indicator system that includes market volatility, MLT market correlation, and price conductivity is proposed. Moreover, the Markov chain Monte Carlo-Diagonalization Method algorithm is proposed to solve the simulation model. The efficacy of the proposed model is validated through a case study of Guangdong electricity market in China. The results demonstrate that the model framework can optimize the most suitable delivery method and MLT percentages based on the development stage. Transitional delivery is most suitable for China's current stage, with composite indicator outperforming physical delivery by 8 % and financial delivery by 5.4 %.

Nomenclature

A. Indices

G	Generation companies	i	Subscript index for generators
L	Medium- and long-term (MLT) market	t	Subscript index for trading hour
D	Day-ahead market	s	Subscript index for scenarios
B. Parameters			
k^{HTP}	Heat-to-power ratio	B^{max}	Upper limit of bid price in electricity market
k^{PAP}	Plant's auxiliary power ratio	p_i^{Gmax}	Upper output limit of generators
M_C, M_{CO_2}	The molar mass of coal and carbon dioxide	p_i^{Gmin}	Lower output limit of generators
LHV	Low heat value of coal	R_i^{up}, R_i^{dn}	Ramping up/down limits of generators
η	Thermal efficiency of power generation	rat	Proportion of MLT contracts
δ_i	Linear coefficient of carbon emission intensity	b_{nm}	Susceptance of line between node n, m

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γ_i	Constant coefficient of carbon emission intensity	$p_{nm,t}^{Lmax}$	Transmission capacity of line between node n and m
a_i, b_i, c_i	Fuel cost coefficient of thermal power generators	M_i^{CBmax}	Upper limit of buying carbon emission allowances (CEAs)
c^{fuel}	Coal cost	M_i^{CBmin}	Lower limit of buying CEAs
$O_{i,t}^L$	MLT contract price	M_i^{CSmax}	Upper limit of selling CEAs
π_i^{SU}	Cost for starting up generators	M_i^{CSmin}	Lower limit of selling CEAs
π_i^{SD}	Cost for shutting down generators	M	Penalty term in Big M method
C. Variables			
p^{GO}	Gross power of generators	β_i	Linear coefficient of bid line
p^N	Net power of generators	$a_{i,t}$	Constant coefficient of bid line
p^{PAP}	Unit's auxiliary power	$\lambda_{(n,i \in \mathcal{W}_n),t}^D$	Day-ahead market clearing price
H	Heat power of generators	λ_t^{carbon}	Carbon market clearing price

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* Corresponding author.

E-mail address: kai.jiang@ncepu.edu.cn (K. Jiang).

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v_c	Coal supply flow rate	$u_{i,t}$	The binary variable indicating running states with 1 and shut-down states with 0
m_{c_s}	The mass of coal and carbon dioxide	A_i^S	Free CEAs allocated by the government
$I_{i,t}$	Carbon emission intensity of generator i	A_i^C	Real CEAs consumed by the GenCo i
f_i^G	Total revenue of Generation company (GenCo) i	$E_{i,t,k}^I$	Strategies sets of GenCo i
$C_{i,t}^L$	GenCo i 's revenue in MLT market	$E_{i,t,k}^B$	Strategies sets of all GenCos except GenCo i
$C_{i,t}^D$	GenCo i 's revenue in day-ahead market	$B_{i,t,k}^{MLT}$	Strategies of GenCo i in MLT market
$C_{i,t}^{fuel}$	Coal cost of GenCo i	$B_{i,t,k}^C$	Strategies of GenCo i in carbon market
$C_{i,t}^{SU}$	Start-up cost of GenCo i	ξ	Convergence factor
$C_{i,t}^{SD}$	Shut-down cost of GenCo i	TF	Transition probability value
$R_{i,t}$	Penalty cost for contract deviation	F^L	Power purchase cost in MLT market
$C_{i,t}^G$	Revenue in carbon market	F^D	Power purchase cost in day-ahead market
$x_{i,t}^D, x_s^F$	Binary variables	F^C	Social welfare in carbon market
$P_{i,t}^L$	Awarded power in MLT market	$\theta_{n,t}$	Voltage angle of bus n at time t
$P_{i,t}^D$	Awarded power in day-ahead market	B_i^{CB}, B_i^{CS}	Price of CEA purchased/sold
$B_{i,t}^D$	Bid strategies in day-ahead market	M_i^{CB}, M_i^{CS}	Amount of CEA purchased/sold
D. Abbreviations			
MLT	Medium- and long-term	MPEC	Mathematical programming with equilibrium constraints
GenCos	Generation companies	EPEC	Equilibrium Problem with Equilibrium Constraints
PTCR	Physical and transitional contracts ratio	MCMC	Markov chain Monte Carlo
CEI	Carbon emission intensity	DM	Diagonalization Method
CEA	Carbon emission allowance	KKT	Karush-Kuhn-Tucker

1. Introduction

China has the world's largest power system, which has supported its rapid economic growth for nearly 30 years [1]. During this period, the power system's dispatch plan used to rely on manual scheduling by the grid company due to historical and technical constraints [2]. To restore the commodity attributes of electricity and enhance the efficiency of resource allocation, China launched its first-round reform of power sector in 2002 [3], when the medium- and long-term (MLT) electricity market was introduced to replace mandatory dispatch plans. Currently, during the ongoing second-round reform, the spot electricity market is gradually being established in China, placing the MLT contract in a controversial transition phase, i.e., from the previous fully-physical delivery towards some form of financial delivery. Recently, China has announced its carbon neutrality goals and officially launched the national carbon market [4], enforcing additional modeling constraints and introducing disturbances to the aforementioned transition. From a global perspective, the simultaneous consideration of decarbonization and marketization in a large-scale power system reform is unprecedented and presents significant challenges.

Currently, during the transition phase towards marketization, the delivery methods of MLT electricity contracts, which integrate the MLT market and spot market, can be categorized into three forms: physical, financial, and transitional contracts. Physical contracts are delivered by imposing constraints to ensure that generation companies (GenCos) meet the electricity volumes stipulated in MLT contracts. In contrast, financial contracts are a type of contract for differences and do not

impose constraints on the market behavior of participants. Transitional contracts, which penalize deviations from the contract, are used in Guangdong Province, China [5]. When the physical and transitional contracts ratio (PTCR) is reduced to zero, the MLT contracts become fully financial delivery. Currently, China mandates that MLT contracts must account for over 90 % of total transaction contracts [6–8], meaning that the PTCR is 90 %, with only 10 % of the contracts being financially executed. The MLT physical and transitional delivery mechanisms, combined with PTCR, form a distinctive linkage between MLT markets and spot market.

Unlike in mature markets such as Pennsylvania-New Jersey-Maryland Interconnection (PJM) and the Nordic power market [9,10], MLT contracts in China are not purely financial agreements. In PJM and Nordic markets, MLT contracts do not constrain system operations or unit dispatch. In contrast, in China, the information from the physically delivered portion of the MLT contracts is integrated into the dispatch plan of the spot market [11]. Furthermore, in developed countries such as Australia [12], the United Kingdom [13], and the United States [14], there is no requirement for MLT contract ratios to exceed a specific threshold, which contrasts with Chinese regulatory practices. From the generation perspective, GenCos need to consider the constraints of MLT contracts in the spot electricity market, where the PTCR of MLT contracts adds nonlinear constraints to the GenCos' decision model. Furthermore, the introduction of the carbon market has increased the complexity of the cost function for GenCos. Thus, it is crucial to establish a simulation model for the electricity and carbon markets and evaluate it to select the most suitable delivery method for MLT contracts in the spot electricity market.

To study the coupling relationship between the forward market and the spot market, scholars have conducted extensive research on the characteristics of electricity market. Some scholars focus on the coupling mechanism between the MLT and spot electricity markets [15–17], some scholars study mathematical models that respond to the coupling relationship between the MLT and spot electricity markets [18,19], and some scholars investigate the impact of policies on the coupling relationship between the MLT and the spot market [20,21]. In a more complex manner, the transition process is accompanied by the launch of the national carbon market, overlaying marketization with decarbonization. Nevertheless, the processes of marketization and decarbonization frequently conflict, resulting in coupling effects between the electricity market and the carbon market. Scholars have found a coupling relationship between the carbon market and the spot electricity market, reflected in price conductivity [22], GenCos' strategies [23], and market clearing outcomes [24,25]. Ref. [26] proposes a multi-stage carbon market clearing mechanism on the assumption that GenCos are not subject to regulation and participate only in the spot electricity market. Ref. [27] develops an electricity market daily clearing model and a carbon market monthly trading model, ignoring the impact of MLT electricity contracts on the electricity and carbon markets. At present, China has not yet achieved the goals of full marketization and decarbonization. **However, most studies are based on the final stage of the dynamic transition process, assuming a fully marketized environment. Existing research has overlooked the dynamic coupling process between marketization and decarbonization.**

To simulate the electricity and carbon markets, a number of studies have been conducted. These studies can be categorized into the following three types based on their methods. **The first type is data-driven methods**, which build models and analyze market behavior by examining extensive historical market data. Ref. [28], based on the historical data from Finnish and German electricity markets, utilizes the extreme learning machine in combination with bootstrap intervals to predict future electricity prices in the two markets. Ref. [29] develops a machine learning framework to predict agents' bids in the electricity market, and the effectiveness of the framework is tested based on actual data from the Australian national electricity market. **The second type is model-driven methods**, which establish mathematical equations or

optimization models based on actual market operating mechanisms or economic theories. Ref. [30] employs computable general equilibrium methods to assess the impact of the national carbon market on the economy, environment, and energy. Ref. [31] develops a mixed-integer nonlinear electricity market clearing model and solves it by the generalized benders decomposition method. In Ref. [32], a bi-level model is built, in which the upper-level model is agents' profits maximization model, and the lower-level model is made up of two market-clearing models: electricity market and carbon market. **The third type is hybrid methods**, which combine data-driven methods and model-driven methods. The market clearing model considering the bidding of multiple strategic GenCos can be better solved by using deep reinforcement learning-based methods [33,34]. Ref. [35] develops a multi-agent reinforcement learning approach to model the electricity market, and uses data from the German wholesale electricity market as a case study. Ref. [36] uses a Takagi-Sugeno-Kang fuzzy inference system for forecasting electricity prices. Ref. [37] utilizes the penalty price factor and threshold to characterize the supply shortage risk punishment and optimizes the strategy of a hybrid power system in the electricity market. **However, the aforementioned simulation method fails to capture the economic and energy flows between the electricity and carbon markets and the GenCos, and does not reflect the dynamic relationships during the transition process.**

To analyze and evaluate the effectiveness of market transitions, it is necessary to establish a comprehensive evaluation indicator system. To achieve this goal, the indicator system can be divided into three parts. **The first part is to identify evaluation indicators.** Ref. [38–40] focuses on economic indicators, analyzing the strategies of GenCos participating in the electricity or hydrogen markets to maximize revenues. Concentrating on the environmental indicators, Ref. [41] introduces a novel indicator called Green Energy to characterize the supply of renewable energy and evaluate the proposed method by analyzing the renewable energy utilization efficiency. Ref. [42] focuses on poverty alleviation indicators, studying the impact of the electricity and carbon markets on poverty reduction. **The second part is to construct a system of indicators.** Ref. [43] establishes two primary indicators: electricity market performance and carbon market performance, which include secondary indicators such as market revenues and carbon emissions. In Ref. [44], a series of consumer-centric indicators are proposed, including service quality, min-max indicator, average efficiency index, and others. Ref. [45] categorizes the indicators into economic and environmental types to evaluate the effectiveness of the proposed hybrid method. **The third part is to determine the weights of the indicators.** Common methods include objective evaluation weighting methods based on historical data, such as principal component analysis [46] and factor analysis [47]; subjective evaluation weighting methods based on expert experience, such as the Delphi method [48] and analytic hierarchy process [49]; and combined subjective and objective evaluation methods, such as combination weighting method [50] and fuzzy comprehensive evaluation method [51]. **However, all of the above studies are based on assumptions of static electricity markets, lacking indicators that reflect the impact of the carbon market on changes in the electricity markets during the transitional process.**

In summary, in the field of modeling electricity and carbon markets, the deregulation process of the electricity market is often neglected, not to mention the overlay of the decarbonization process on this basis. During the transition process, the coupling flows between the electricity market and the carbon market, the strategic behavior of market participants, and the impact of dynamic carbon emission intensity (CEI) on the electricity and carbon markets are rarely considered. Therefore, based on the previous research, the paper proposes a bi-level model framework that incorporates the participation of multiple GenCos in the marketization of the electricity market, while also considering the implementation of the carbon market. The simulation model is then analyzed and evaluated using Guangdong electricity market as a case

study.

The key contributions are listed as follows.

- (1) For the transitional process of the electricity market driven by deregulation and decarbonization, the deregulation process is innovatively characterized by changes in the delivery methods of MLT contracts, categorized into physical delivery, transitional delivery, and financial delivery. The decarbonization process is analyzed through a bottom-up approach by constructing a secondary carbon market trading model. Taking into account the developmental stages of the power system, a bi-level game model framework for the electricity and carbon markets has been developed.
- (2) For GenCos participating in multi-market trading across multiple timescales, a strategic bidding model for individual GenCo and an equilibrium model for the inter-game of multiple GenCos have been established. To address the equilibrium problem with equilibrium constraints (EPEC) posed by this model, the Markov chain Monte Carlo (MCMC)-diagonalization method (DM) algorithm is utilized, which integrates individual optimal strategies with group Nash equilibrium, thereby enhancing the solution's effectiveness.
- (3) To comprehensively assess the impacts of multiple markets across various contract delivery methods, an indicator system is proposed to evaluate electricity and carbon markets during the transitional process. Building on common indicators from previous research, this paper introduces market volatility, MLT market correlation, and price conductivity, and employs the analytic hierarchy process (AHP) to reflect changes in market revenues, risks, and the interplay between electricity and carbon markets during the transitional process.

The rest of this paper is organized as follows. Section 2 introduces the framework of the bi-level model of electricity and carbon markets. Section 3 formulates the upper-level and the lower-level models, and reformulates the bi-level model as an EPEC model. Model linearization methods and algorithms are described in Section 4. In Section 5, this paper examines three cases under different MLT electricity contract delivery methods and analyzes key factors. Related conclusions are presented in Section 6.

2. Model framework and assumptions

In this section, a simulation model framework for GenCos involved in electricity and carbon markets trading is established. Subsequently, the assumptions for the electricity and carbon markets based on this framework are introduced.

2.1. Model framework

As shown in Fig. 1, the upper-level model represents GenCos' strategic decision-making models, while the lower-level model consists of MLT electricity market, spot electricity market, and carbon market. At the upper level, multiple GenCos participate in bidding to maximize their respective profits, engaging in a non-cooperative game with one another. The GenCos sequentially participate in the MLT market, the spot market, and the carbon market at the lower level based on the market organization. Physical, economic, and information flows are transmitted between the upper-level and lower-level models. The specific trading process unfolds as follows: first, GenCos submit bidding information to the independent system operator (ISO) or carbon trading center. These institutions are responsible for verifying the bids by transaction constraints. After verification, the transaction information is transmitted to the lower level for market clearing. Subsequently, the clearing results are communicated back to the relevant GenCos, which adjust their dispatch plans accordingly. The optimization results of the

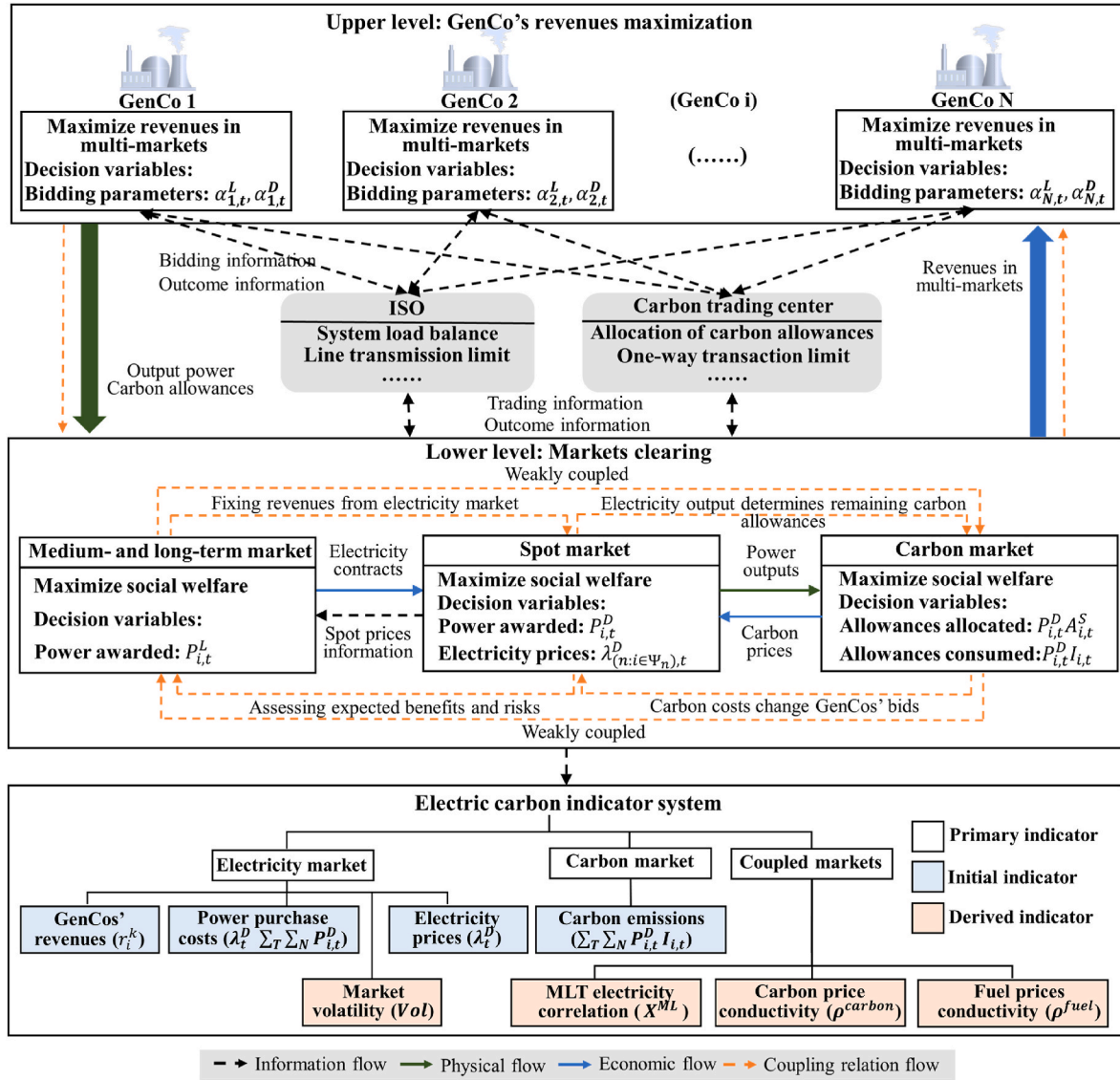


Fig. 1. Model framework.

upper and lower-level models are interdependent, indicating a tightly coupled relationship.

The decision variables in the MLT market, spot market, and carbon market models function as constraint constants for each other. From the perspective of a trading process, to mitigate risks in the spot market, GenCos and users sign large quantities of electricity contracts for future periods in the MLT electricity market [52]. Meanwhile, historical spot electricity price information serves as a price reference for MLT transactions. The spot electricity market clearing outcomes determine the output of thermal units, and the resulting carbon emissions affect the supply and demand of allowances in the carbon market. Carbon market prices impact the generation costs of thermal GenCos, which, in turn, affect the clearing order in the spot market. Although the MLT electricity market and the carbon market are not directly coupled, they are indirectly linked through the spot market.

To evaluate the simulation results, an indicator system comprising three primary indicators and eight secondary indicators is developed. The initial indicators are expressed in Section 5.2 and the derived indicators are expressed in Sections 5.3 - 5.6. Using the AHP, comprehensive evaluation results for the eight secondary indicators are obtained, enabling a comparison to determine the most suitable contract delivery method for China at the current stage.

2.2. Electricity market assumptions

The Chinese electricity market operates as a centralized electricity market. The ISO organizes the annual MLT transactions for the following year at the end of each year. GenCos and users trade power and allocate it proportionally to each month in the annual MLT electricity market, with the majority of MLT trading occurring through bilateral negotiations. At the end of each month, GenCos trade the next month's power in the monthly market and allocate it proportionally to each day. In the spot electricity market, GenCos need to consider the delivery of MLT contracts, with all electricity traded through centralized bidding. Based on the electricity market trading process and mechanisms, four assumptions are made in this paper.

- 1) MLT transactions are conducted in batches on annual, monthly, and weekly schedules. Given that the main focus is on the economic dispatch analysis of the spot market, multiple MLT markets can be regarded as equivalent to a single one.
- 2) In the paper, the spot market refers to day-ahead market. The clearing results of the day-ahead market serve as both a settlement basis and a pre-dispatch plan for the system operator. Meanwhile, power mismatches between day-ahead market awards and real-time

output are regulated and penalized. In practice, the differences between the dispatch results of the day-ahead market and the real-time market are minimal.

- 3) The electricity market simulation model adopts the DC power flow model, with the analysis time scale set to hourly intervals.
- 4) It is assumed that only the GenCos participate in the bidding process in this paper. The current approach to centralized bidding is based on two distinct methods. In the first method, GenCos participate in the bidding process, while users simply declare their demand capacity. In the second method, both GenCos and users participate in the bidding process. Ideally, these two methods should not influence the clearing outcome.

2.3. Carbon market assumptions

In China, GenCos primarily dominate the carbon market, accounting for about 45 % of the nation's carbon emissions [53]. Unlike carbon markets in other countries [54–57], the Chinese national carbon market operates as an intensity-based system, where carbon emission allowances are distributed free of charge [58]. The carbon emission allowances (CEAs) are allocated to GenCos in the primary market using the output-based allocation method [59]. In the model framework of this paper, units with actual emission intensities lower than the standard intensity are granted additional allowances, which can be sold. Conversely, GenCos exceeding the standard intensity must purchase additional allowances to compensate for the excess emissions. Each allowance offsets one ton of carbon dioxide emissions. Allowances can be traded daily in the secondary market, with reconciliations conducted annually [60].

- 1) Operating on a daily time scale, the carbon market assumes daily surpluses or deficits of CEAs are balanced to meet long-term equilibrium targets. Due to the annual settlement cycle of the carbon market, which differs from the daily settlement of the electricity market, GenCos are assumed to adopt a daily settlement strategy. This approach helps avoid penalties for insufficient CEAs at the end of the year.
- 2) The carbon market is assumed to consist of both the power industry and other sectors. Under this assumption, the carbon price is less likely to fluctuate due to changes in the trading behavior of GenCos.

3. Model formulation

In this section, a bi-level model for electricity and carbon markets is formulated. The upper-level model represents GenCos' strategic decision-making model. The decisions of multiple GenCos interact with each other and constitute a complex strategy space. This paper introduces the MCMC algorithm to sample strategy combinations among GenCos and proposes a transfer probability (TP) function to model their decision processes. The lower-level models consist of market clearing models, where market clearing occurs sequentially.

3.1. Dynamic CEI model

In this part, the relationship between CEI and power output is analyzed and modeled. Three theorems relevant to the derivation process are summarized below [31].

- 1) Theorem 1: Boiler efficiency is positively correlated with plant load factor (PAF) [61].
- 2) Theorem 2: Plant's auxiliary power (PAP) is negatively correlated with PAF [62,63].
- 3) Theorem 3: Coal consumption rate (CCR) of the plant is negatively correlated with PAP [64].

$$P^{GO} \cdot k^{HTP} = H \quad (1)$$

$$P^{GO} = P^N + P^{PAP} \quad (2)$$

$$P^{GO} \cdot k^{PAP} = P^{PAP} \quad (3)$$

$$m_c = v_c \cdot \Delta t \quad (4)$$

$$m_{CO_2} = v_c \cdot \Delta t \cdot M_{CO_2} / M_C \quad (5)$$

As shown in Eq. (1), a generator is capable of producing heat power. The ratio that converts heat into power is typically fixed at a specific value [65]. Eq. (2) represents that the electricity produced by generator can be subdivided into net power and PAP. Eq. (3) shows the relationship between gross electricity and PAP. Eqs. (4) and (5) show the mass of coal and carbon dioxide.

$$(P^{GO} + H) \cdot \Delta t = \eta \cdot LHV \cdot m_c \quad (6)$$

$$P^N = \frac{1 - k^{PAP}}{1 + k^{HTP}} \cdot \eta \cdot LHV \cdot m_c \quad (7)$$

$$I_{i,t} = \frac{m_{CO_2}}{1000 P^N \cdot \Delta t} = \frac{M_{CO_2} \cdot (1 + k^{HTP})}{1000 M_C \cdot \eta \cdot (1 - k^{PAP})} \quad (8)$$

$$I_{i,t} = -\delta_i P_{i,t}^D + \gamma_i \quad (9)$$

The relationship between coal consumption and unit output is obtained according to the conservation of energy, as shown in Eq. (6). The net power is derived from Eqs. (1)–(6), as shown in Eq. (7). The CEI can be represented by Eq. (8). The thermal efficiency η and the plant's auxiliary power ratio k^{PAP} are negatively and positively correlated with the CEI respectively. It can be observed that CEI is negatively correlated with net power. The field data sampled by the continuous emission monitoring system (CEMS) installed in China are shown in Fig. 2. The negative correlation between dynamic CEI and output is expressed as a linear function to simplify the modeling process and facilitate the solution, as shown in Eq. (9).

3.2. Upper-level: GenCo's revenue maximization

The objective function of the upper-level model is to maximize GenCo's revenues, which are derived from the MLT electricity market, the spot electricity market, and the carbon market, as depicted in Eq. (10).

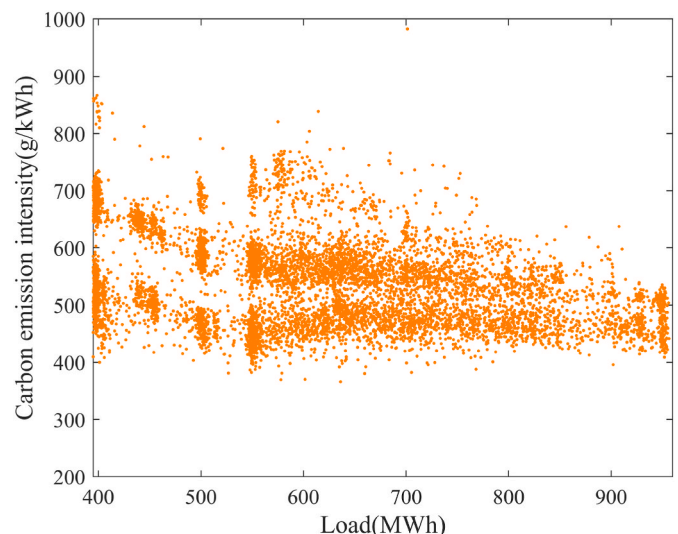


Fig. 2. Carbon emission intensity data.

$$\max f_i^G = \sum_{t \in \Omega^T} \left(C_{i,t}^L + \sum_{s \in \Omega^S} x_s^D C_{i,t}^D - C_{i,t}^{fuel} - (C_{i,t}^{SU} + C_{i,t}^{SD}) \right) - \sum_{s \in \Omega^S} x_s^E \sum_{t \in \Omega^T} R_{i,t} + \sum_{t \in \Omega^T} C_{i,t}^G \quad (10)$$

$$C_{i,t}^L = P_{i,t}^L O_{i,t}^L \quad (11)$$

$$C_{i,t}^D = (P_{i,t}^D - P_{i,t}^L) \cdot \lambda_{(n,i \in \Psi_n),t}^D \quad (12)$$

$$C_{i,t}^{fuel} = a_i P_{i,t}^{D^2} + b_i P_{i,t}^D + c_i \quad (13)$$

$$C_{i,t}^{SU} = (u_{i,t} - u_{i,t-1}) \cdot u_{i,t} \cdot \pi_i^{SU} \quad (14)$$

$$C_{i,t}^{SD} = (u_{i,t} - u_{i,t-1}) \cdot u_{i,t} \cdot \pi_i^{SD} \quad (15)$$

$$R_{i,t} = (0.9 P_{i,t}^D - P_{i,t}^L) \cdot 2 \cdot \lambda_{(n,i \in \Psi_n),t}^D \cdot P_{i,t}^D \quad (16)$$

$$C_{i,t}^G = \lambda_t^{carbon} (A_i^S - (\delta_i P_{i,t}^D + \gamma_i)) \cdot P_{i,t}^D \quad (17)$$

Eq. (10) represents the objective function of the GenCos. The revenues from MLT electricity market and spot electricity market are given by Eqs. (11) and (12). Eq. (13) indicates the fuel cost of the GenCo. The shut down/start up costs are shown in Eqs. (14) and (15). Eq. (16) represents the penalty costs incurred during the transitional delivery of the MLT electricity contracts. Eq. (17) shows the revenues from carbon markets.

The decision variable of the upper-level model is the bid parameter α of the GenCo, and the constraint is shown in Eq. (18).

$$0 \leq B_{i,t}^D \leq B^{max} \quad (18)$$

$$B_{i,t}^D = \beta_i P_{i,t}^D + \alpha_{i,t} \quad (19)$$

$$\alpha_{i,t} = \alpha_{1,i,t} C_{i,t}^{fuel} + \alpha_{2,i,t} \quad (20)$$

$$E_{i,t,k}^L = \{B_{i,t,k}^{MLT}, B_{i,t,k}^D, B_{i,t,k}^C\} \quad (21)$$

$$E_{i,t,k}^B = \{B_{j,t,k}^{MLT}, B_{j,t,k}^D, B_{j,t,k}^C \mid j \neq i\} \quad (22)$$

In this paper, the bid model of GenCo is constructed by using the supply function equilibrium (SFE) model [66]. GenCos submit a diagonal line to the ISO, which responds to the power generation cost, as depicted in Eq. (19). Eq. (19) has the same functional form as the derivation of Eq. (2), where $\beta \approx 2a$. In the paper, it is assumed that the GenCos engage in the market through a game involving the parameter α , and the parameter β is regarded as public market information. This assumption is corresponded with the bid- α game [67]. Moreover, α can be decomposed into the form of Eq. (20) with excellent scalability. Eq. (21) represents the strategy sets of GenCo i . Eq. (22) represents the strategy sets of other GenCos.

The MCMC algorithm is a numerical computation and simulation method that is employed in a wide range of applications, including image segmentation [68], testing in epidemic models [69], and artificial intelligence [70]. Some of the most commonly employed MCMC algorithms include the Metropolis-Hastings algorithm, the Gibbs algorithm and Hamiltonian Monte Carlo, and others. Direct sampling is challenging to address the distribution of multi-dimensional distributions. The MCMC algorithm constructs a Markov chain, from which samples are taken from the Markov chain to approximate the target distribution. Similarly, the strategy space of multi-agents is also a multi-dimensional distribution problem. The MCMC algorithm can be used to sample from a diversity of strategy distributions, which can effectively obtain an

approximate optimal solution to the complex problem [71].

In the MCMC algorithm, the bidding strategy of each GenCo is a state within the system, and the set of strategies of all GenCos constitutes the state space. In the current state, alternative states are generated by the addition of Gaussian noise to the strategy. Subsequently, the TP function is employed to determine whether the alternative state should be accepted or rejected. By demonstrating that the TP function satisfies the *detailed balance condition*, it can be guaranteed that the Markov chain will eventually converge to the target distribution. This implies the existence of a locally or globally optimal solution. The TP function represents the most critical component in the upper-level model based on the MCMC algorithm. The TP function of the GenCo is defined as Eq. (23):

$$TF = \min \left(1, \exp \left(- \frac{f_i^G(E_{i,t,k}^L, E_{i,t,k}^B) - f_i^G(E_{i,t,k+1}^L, E_{i,t,k+1}^B)}{\xi} \right) \right) \quad (23)$$

The parameter ξ represents a convergence factor. When ξ is large, the probability of accepting a suboptimal strategy is increased, and the likelihood of falling into a local optimum is reduced. Conversely, when ξ is small, the model converges more rapidly. A variable $u \sim U(0, 1)$ is generated randomly, and if the TF is greater than u , the alternative strategy is accepted. Conversely, if the TF is less than u , the current strategy is retained. The detailed algorithm for GenCo's decision-making process based on MCMC algorithm is presented in **Algorithm 1**.

Algorithm 1. Metropolis-Hastings algorithm

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1: INITIALIZE: Markov chain state  $E_{i,t,0}^L$ 
2: WHILE iteration index  $k \leq K$ 
3:   Generate Gaussian random noise  $\kappa$ 
4:   Update alternative state  $E_{i,t,k+1}^L = E_{i,t,k}^L + \kappa$ 
5:   Calculate the transfer probability  $TF$  by Eq. (22)
6:   Generate random variable  $u \sim U(0, 1)$ 
7:   IF  $TF > u$ 
8:     Update state  $E_{i,t,k+1}^L = E_{i,t,k+1}^L$ 
9:   ELSE
10:    Update state  $E_{i,t,k+1}^L = E_{i,t,k}^L$ 
11:  END
12: END WHILE

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3.3. Lower-level-1: medium- and long-term electricity market clearing

The objective function of the MLT electricity market clearing model is to maximize social welfare, which in this paper is expressed as minimizing the cost of electricity consumption, as illustrated in Eq. (24).

$$\min F^L = \sum_{i=1}^N \sum_{t=1}^T (\beta_i P_{i,t}^L + \alpha_{i,t}) P_{i,t}^L \quad (24)$$

The decision variable for the MLT electricity market clearing model is denoted by $P_{i,t}^L$. The constraints are presented in (25)–(29).

$$P_i^{Gmin} \leq P_{i,t}^L \leq P_i^{Gmax} \quad (25)$$

$$P_{i,t+1}^L - P_{i,t}^L \leq R_i^{up} \quad (26)$$

$$P_{i,t}^L - P_{i,t+1}^L \leq R_i^{dn} \quad (27)$$

$$\sum_{i=1}^N P_{i,t}^L = P_t^{Lall} \quad (28)$$

$$\sum_{t=1}^T \sum_{i=1}^N P_{i,t}^L = rat \times \sum_{t=1}^T D_t \quad (29)$$

Eq. (25) represents the output limits of units. The ramping up/down capacity limits are shown in Eqs. (26) and (27). Eq. (28) specifies that

the total traded power from all units during a given time period must not surpass the power outlined in the planned curve for that same period. Eq. (29) indicates that the total amount of electricity traded on the MLT during each time period must be in accordance with the policy's requirements, expressed as a proportion of spot market trading.

3.4. Lower-level-2: spot electricity market clearing

The objective function of the spot electricity market clearing model aims to maximize social welfare, which is equivalent to minimizing the cost of electricity consumption, as shown in Eq. (30).

$$\min F^D = \sum_{i=1}^N \sum_{t=1}^T (\beta_{i,t} P_{i,t}^D + \alpha_{i,t}) P_{i,t}^D \quad (30)$$

The decision variables for the spot electricity market clearing model are $P_{i,t}^D$ and $\theta_{n,t}$, with the constraints are presented in Eqs. (31)–(38).

$$P_i^{Gmin} \leq P_{i,t}^D \leq P_i^{Gmax} [\mu_{i,t}^{D,min}, \mu_{i,t}^{D,max}] \quad (31)$$

$$P_{i,t+1}^D - P_{i,t}^D \leq R_i^{up} [v_{i,t}^{D,up}] \quad (32)$$

$$P_{i,t}^D - P_{i,t+1}^D \leq R_i^{dn} [v_{i,t}^{D,dn}] \quad (33)$$

$$\sum_{i \in \Psi_n} P_{i,t}^D + \sum_{m \in \Psi_n} b_{nm} (\theta_{n,t} - \theta_{m,t}) - \sum_{i \in \Psi_n} P_{i,t}^D = 0 [\lambda_t^D] \quad (34)$$

$$b_{nm} (\theta_{n,t} - \theta_{m,t}) \leq P_{nm,t}^{Lmax} [l_{n,t}^{D,+}] \quad (35)$$

$$-b_{nm} (\theta_{n,t} - \theta_{m,t}) \leq P_{nm,t}^{Lmax} [l_{n,t}^{D,-}] \quad (36)$$

$$-\pi \leq \theta_{n,t} \leq \pi [\theta_{n,t}^{D,min}, \theta_{n,t}^{D,max}] \quad (37)$$

$$\theta_{1,t} = 0 [\theta_{1,t}^1] \quad (38)$$

Eq. (31) represents the output limits of units. The ramping up/down capacity limits are shown in Eqs. (32) and (33). Eq. (34) represents the node power balance. The transmission line capacity limits are shown in Eqs. (35) and (36). Eq. (37) represents the node phase angle limits. Eq. (38) represents the balancing node.

3.5. Lower-level-3: carbon market clearing

The objective function of the carbon market clearing model aims to maximize social welfare, as shown in Eq. (39).

$$\max F^C = \sum_{i=1}^N B_i^{CB} M_i^{CB} - B_i^{CS} M_i^{CS} \quad (39)$$

$$A_i^S = P_{i,t}^D I_{i,t}^S \quad (40)$$

$$A_i^C = \delta_i P_{i,t}^{D^2} + \gamma_i P_{i,t}^D \quad (41)$$

$$\sum_{i=1}^N M_i^{CB} = \sum_{i=1}^N M_i^{CS} [\lambda_t^{carbon}] \quad (42)$$

$$M_i^{CBmin} \leq M_i^{CB} \leq M_i^{CBmax} [\tau_i^{B,min}, \tau_i^{B,max}] \quad (43)$$

$$M_i^{CSmin} \leq M_i^{CS} \leq M_i^{CSmax} [\tau_i^{S,min}, \tau_i^{S,max}] \quad (44)$$

Eq. (40) represents the CEAs allocated in the primary carbon market, and the CEAs consumed by GenCos is shown in Eq. (41). Eq. (42) represents the CEAs balance constraint. Eqs. (43) and (44) represent the available CEAs of GenCos.

3.6. Multi-markets equilibrium model considering multi-strategic agents

The model of a single GenCo making decisions in a multi-market is a bi-level model with the upper leader being the strategic GenCo and the lower follower being the market, and the upper-level model is shown in Eqs. (45)–(47):

$$\max f_i(b_i, y^*) \quad (45)$$

$$s.t. g_i(b_i, y^*) \leq 0 \quad (46)$$

$$h_i(b_i, y^*) = 0 \quad (47)$$

Where, f_i represents the decision function of GenCo i , b_i is the decision variable of the upper-level model, y^* is the optimal solution of the lower-level model, and g_i and h_i are inequality and equation constraints of upper model respectively.

The lower-level model is shown in Eqs. (48)–(50):

$$\min F(y, b_i^*) \quad (48)$$

$$s.t. G(y, b_i^*) \leq 0 [\mu] \quad (49)$$

$$H(y, b_i^*) = 0 [\sigma] \quad (50)$$

Where, F is the decision function for market clearing, y is the decision variable of the lower-level model, b_i^* is the optimal solution of the upper-level model, and G and H are inequality and equation constraints of upper-level model respectively.

When the strategies of the other GenCos are fixed, the constraints of the lower-level model are transformed into the upper-level model through the Karush-Kuhn-Tucker (KKT) theorem [72]. The bi-level model of a single strategic GenCo's participation in the market is transformed into the Mathematical programming with equilibrium constraints (MPEC) model [73], as shown in Eqs. (51)–(57):

$$\max f_i(b_i, y^*) \quad (51)$$

$$s.t. g_i(b_i, y^*) \leq 0 \quad (52)$$

$$h_i(b_i, y^*) = 0 \quad (53)$$

$$G(y, b_i^*) \leq 0 \quad (54)$$

$$H(y, b_i^*) = 0 \quad (55)$$

$$\nabla_y F(y, b_i^*) + \mu \nabla_y G(y, b_i^*) + \sigma \nabla_y H(y, b_i^*) = 0 \quad (56)$$

$$0 \leq G(y, b_i^*) \perp \mu \geq 0 \quad (57)$$

Where, μ and σ represent the Lagrange multiplier.

The optimal decision-making model for each GenCo is an MPEC model, and when multiple strategic GenCos each make decisions in the market to maximize their revenue, the model that solves the equilibrium of multiple MPEC models constitutes an EPEC model.

4. Solution methodology

In this section, the structure of an innovative algorithm that solves bi-level game-based model for electricity and carbon markets is promoted, as shown in Fig. 3. The KKT theorem is initially employed to convert the lower-level model into the optimality condition of the upper-level model [72]. Subsequently, the transformed model is linearized to convert the MPEC of an individual GenCo in electricity and carbon markets into a linear model. All GenCos' decision-making process is modeled as EPEC. The EPEC is solved by MCMC-DM algorithm, which is a combination of the MCMC algorithm and DM algorithm.

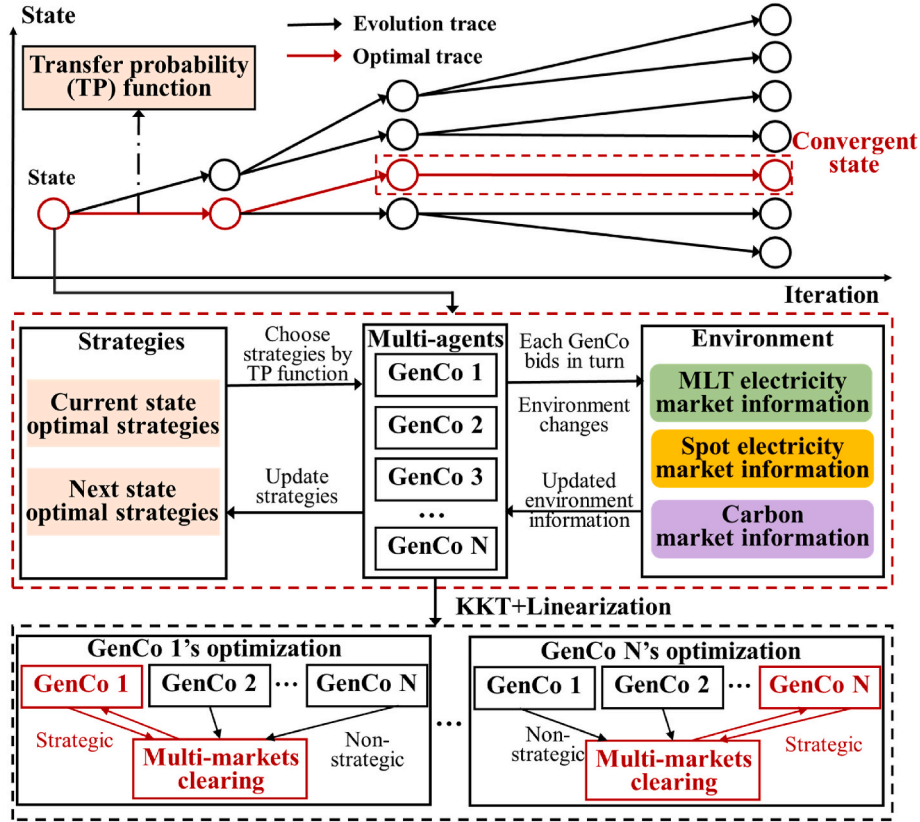


Fig. 3. Algorithm structure.

4.1. Model reformulation and linearization

The lower-level market clearing model is transformed by using the KKT theorem. The resulting lower-level model is shown in Eqs. (58)–(76):

$$\beta_{i,t} - \lambda_{(n,i \in \Psi_n)} - \mu_{i,t}^{D,min} + \mu_{i,t}^{D,max} - v_{i,t}^{D,up} + v_{i,t}^{D,dn} = 0 \quad \forall i, t = 1 \quad (58)$$

$$\beta_{i,t} - \lambda_{(n,i \in \Psi_n)} - \mu_{i,t}^{D,min} + \mu_{i,t}^{D,max} - v_{i,t-1}^{D,up} + v_{i,t-1}^{D,dn} = 0 \quad \forall i, t = T \quad (59)$$

$$\beta_{i,t} - \lambda_{(n,i \in \Psi_n)} - \mu_{i,t}^{D,min} + \mu_{i,t}^{D,max} - v_{i,t-1}^{D,up} + v_{i,t}^{D,dn} = 0 \quad \forall i, t \in (1, T) \quad (60)$$

$$\sum_{m \in \Psi_n} b_{nm} (\lambda_{n,t} - \lambda_{m,t}) - \vartheta_{n,t}^{D,min} + \vartheta_{n,t}^{D,max} + \sum_{m \in \Psi_n} b_{nm} (l_{n,t}^{D,+} - l_{n,t}^{D,-}) + \vartheta_{1,t} = 0 \quad \forall t, n = 1 \quad (61)$$

$$\sum_{m \in \Psi_n} b_{nm} (\lambda_{n,t} - \lambda_{m,t}) - \vartheta_{n,t}^{D,min} + \vartheta_{n,t}^{D,max} + \sum_{m \in \Psi_n} b_{nm} (l_{n,t}^{D,+} - l_{n,t}^{D,-}) = 0 \quad \forall t, n > 1 \quad (62)$$

$$0 \leq (p_i^{G,max} - p_{i,t}^D) \perp \mu_{i,t}^{D,max} \geq 0 \quad \forall i, t \quad (63)$$

$$0 \leq (p_{i,t}^D - p_i^{G,min}) \perp \mu_{i,t}^{D,min} \geq 0 \quad \forall i, t \quad (64)$$

$$0 \leq (p_{i,t}^D - p_{i,t+1}^D + R_i^{up}) \perp v_{i,t}^{D,up} \geq 0 \quad \forall i, t < T \quad (65)$$

$$0 \leq (p_{i,t+1}^D - p_{i,t}^D + R_i^{dn}) \perp v_{i,t}^{D,dn} \geq 0 \quad \forall i, t < T \quad (66)$$

$$0 \leq (\pi - \theta_{n,t}^D) \perp \vartheta_{n,t}^{D,max} \geq 0 \quad \forall n, t \quad (67)$$

$$0 \leq (\pi + \theta_{n,t}^D) \perp \vartheta_{n,t}^{D,min} \geq 0 \quad \forall n, t \quad (68)$$

$$0 \leq (P_{nm,t}^{L,max} - b_{nm} (\theta_{n,t}^D - \theta_{m,t}^D)) \perp l_{i,t}^+ \geq 0 \quad (69)$$

$$0 \leq (P_{nm,t}^{L,max} + b_{nm} (\theta_{n,t}^D - \theta_{m,t}^D)) \perp l_{i,t}^- \geq 0 \quad (70)$$

$$B_i^{CS} - \tau_i^{S,min} + \tau_i^{S,max} - \lambda^C = 0 \quad \forall i \quad (71)$$

$$-B_i^{CB} - \tau_i^{B,min} + \tau_i^{B,max} + \lambda^C = 0 \quad \forall i \quad (72)$$

$$0 \leq (M_i^{CB} - M_i^{CB,min}) \perp \tau_i^{B,min} \geq 0 \quad \forall i \quad (73)$$

$$0 \leq (M_i^{CB,max} - M_i^{CB}) \perp \tau_i^{B,max} \geq 0 \quad \forall i \quad (74)$$

$$0 \leq (M_i^{CS} - M_i^{CS,min}) \perp \tau_i^{S,min} \geq 0 \quad \forall i \quad (75)$$

$$0 \leq (M_i^{CS,max} - M_i^{CS}) \perp \tau_i^{S,max} \geq 0 \quad \forall i \quad (76)$$

In this part, the big M method is used to linearize the complementary relaxation conditions (63)–(70), (73)–(76) [72]. Taking Eq. (63) for example, the linearized formulas are shown in Eqs. (77) and (78):

$$0 \leq \mu_{i,t}^{D,max} \leq x_{i,t}^{0-1} M \quad \forall i, t \quad (77)$$

$$0 \leq (p_i^{G,max} - p_{i,t}^D) \leq M (1 - x_{i,t}^{0-1}) \quad \forall i, t \quad (78)$$

The binary expansion method is used to linearize the portion of the model where the two decision variables are multiplied together [74]. Taking $p_{i,t}^D \lambda_{(n,i \in \Psi_n),t}$ for example, the expansion takes the following form of Eq. (79).

$$P_{i,t}^D \lambda_{(n,i \in \mathcal{V}_n),t} = P_{i,t}^D \lambda_{i,t}^{min} + P_{i,t}^D \Delta \lambda^D \sum_{q=0}^{Q-1} 2^q x_{i,t}^{D,0-1} \quad (79)$$

Assuming that $\xi_{i,t}^D = P_{i,t}^D x_{i,t}^{D,0-1}$, Eq. (79) can be expressed in the form of Eq. (80). Subsequently, $P_{i,t}^D \lambda_{(n,i \in \mathcal{V}_n),t}$ can be linearized using the big M method, as shown in Eqs. (81) and (82).

$$P_{i,t}^D \lambda_{(n,i \in \mathcal{V}_n),t} = P_{i,t}^D \lambda_{i,t}^{min} + \xi_{i,t}^D \Delta \lambda^D \sum_{q=0}^{Q-1} 2^q \quad (80)$$

$$0 \leq P_{i,t}^D - \xi_{i,t}^D \leq M(1 - x_{i,t}^{D,0-1}) \quad \forall i, t \quad (81)$$

$$0 \leq \xi_{i,t}^D \leq x_{i,t}^{D,0-1} M \quad \forall i, t \quad (82)$$

4.2. MCMC-DM algorithm

The DM algorithm, also known as the fixed-point iteration, is employed to address problems with multiple variables that are interrelated. In each iteration of the DM algorithm, one variable is optimized, the other variables are fixed, and the process is repeated until the solution for all variables no longer changes [75]. However, the DM algorithm does not always find a globally optimal solution. The MCMC algorithm's capacity to sample the entire solution space precludes the solution from becoming trapped in a local optimum. This paper presents a novel algorithm to solve the global approximate optimal solution of the EPEC model with multiple strategic GenCos participating in the markets. The method in this article combines the DM with the MCMC algorithm, offering a robust and efficient solution to this complex problem.

The steps of MCMC-DM algorithm are outlined as follows.

Step 1: Provide data on generator parameters, grid parameters, market parameters, load parameters, etc. Input the initial strategies of each GenCo.

Step 2: The strategy of the GenCo i in the k th round of offers is $B_i^{(k)}$. Fix the strategies of other GenCos, i.e., $B_{-i}^{(k)} = \{B_1^{(k-1)*}, \dots, B_{i-1}^{(k-1)*}, B_{i+1}^{(k-1)*}, \dots, B_N^{(k-1)*}\}$. Then, determine the optimal strategies of GenCo i in generation k and implement this as an alternative state.

Step 3: Calculate the transfer probability TF , and generate random variable $u \sim U(0, 1)$.

Step 4: Compare TF and u . If $TF > u$, the next state will be the alternative state developed in step 2. Otherwise, the current state will be retained.

Table 1
Basic data for units in different groups.

	Group1	Group2	Group3
Capacity (MW)	[400, 735]	[300, 1000]	[300, 1000]
P_i^{Gmin} / Capacity	[0.2, 0.4]	[0.4, 0.6]	[0.4, 0.6]
P_i^{Gmax} / Capacity	1	1	1
a_i (CNY/MWh ²)	0.14	[0.1, 0.15]	[0.1, 0.15]
b_i (CNY/MWh)	[400, 600]	[200, 300]	[300, 600]
β_i (CNY/MWh ²)	0.28	[0.2, 0.3]	[0.2, 0.3]
$\alpha_{1,i}$ (CNY/t)	[0.5, 0.75]	[0.25, 0.4]	[0.4, 0.75]
π_i^{SU}, π_i^{SD} (CNY)	50000	[150000, 200000]	[150000, 200000]
Unit number	22	18	40
Fuel type	Gas	Coal	Coal
GenCos' ID	{22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32}	{1, 2, 3, 5, 9, 10, 15, 36, 39}	{4, 6, 7, 8, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 33, 34, 35, 37, 38, 40}

Step 5: Once the optimal strategies for generation k of GenCo i have been determined, steps 2–4 are repeated until all N GenCos' optimal strategies for the k th round have been identified.

Step 6: Continue repeating Steps 2 through 5 until each GenCo's revenue difference Δr_i^k is less than convergence range ε .

The detailed algorithm of MCMC-DM optimization is shown in **Algorithm 2**.

Algorithm 2. MCMC-DM optimization

```

1: Input: M-L market clearing output, load, unit parameters, etc.
2: Initialize the iteration index  $k = 0$ , all GenCos' strategies  $\{B_1^k, B_2^k, \dots, B_N^k\}$ , all GenCos' profits  $\{r_1^k, r_2^k, \dots, r_N^k\}$ 
3: WHILE iteration index  $k \leq K$ 
4:   WHILE GenCo index  $i \leq N$ 
5:     Optimize the GenCo  $i$ 's strategies over the other GenCos' fixed strategies
6:     Update alternative state in Algorithm 1 with the strategies in step 5
7:     Calculate the transfer probability  $TF$  by Eq. (22)
8:     Generate random variable  $u \sim U(0, 1)$ 
9:     IF  $TF > u$ 
10:      Update optimal strategies  $B_i^{op} = B_i^k$ 
11:    ELSE
12:      Update optimal strategies  $B_i^{op} = B_i^{op}$ 
13:    END IF
14:  END WHILE
15:  IF Each GenCo's revenue difference  $\Delta r_i^k$  is less than convergence range  $\varepsilon$ 
16:    BREAK
17:  END
18: END WHILE

```

5. Case study

In this section, case studies are conducted using MATLAB R2021b and GUROBI 11.0 on a computer equipped with an Intel Core i7-12700H CPU (2.3 GHz) and 16 GB of RAM. Results in 3 case studies based on Guangdong electricity market are analyzed, focusing on the physical, transitional, and financial delivery methods of contracts. Based on this, 4 scenarios are analyzed: 1) In the "DEC" scenario, the decomposition curve and composition of MLT contracts change; 2) In the "PER" scenario, the percentage of MLT contracts changes; 3) In the "FP" scenario, the fuel prices change; 4) In the "CP" scenario, the carbon prices change.

5.1. Base data

The electricity system consists of 3 different GenCos groups with about 80 units. The fuel cost coefficient a_i is randomly set between 0.1 and 0.15 CNY/MWh², while b_i is set between 200 and 600 CNY/MWh [76]. The linear coefficient β_i of GenCos' bid curve is set between 0.2 and 0.3 CNY/MWh². The constant coefficient $\alpha_{1,i}$ of GenCos' bid curve is set between 0.25 and 0.75 CNY/t, while $\alpha_{2,i}$ is determined by GenCos' strategies. Together, they constitute α_i in bid- α game. The minimum output P_i^{Gmin} is randomly selected between 20% and 60 % of the unit's capacity, the maximum output P_i^{Gmax} is equal to the unit's capacity. Based on this, the parameters vary across different GenCo groups, as shown in Table 1. Table 2 presents the relationship between the parameters of dynamic CEI and unit output, as derived from CEMS data collected from Chinese power plants. It is assumed that there are 40 GenCos, each operating two units with identical parameters. The bidding price limit of electricity market is set between 0 and 1500 CNY/MWh. The coal price is 800CNY/ton and the gas price is 2.2 CNY/m³ [77,78].

MLT decomposition curves include uniform curve, peak to valley curve, and customized curve. Uniform curve refers to the decomposition of MLT contracts into each period uniformly. Peak to valley curve is decomposed proportionally according to the peak and valley periods, and customized curve is decomposed proportionally according to the

Table 2
Regression analysis of dynamic CEI.

Fuel type	Benchmark (g/kWh)	Linear coefficient (g/GWh)	Constant term (g/kWh)
Coal	800	-0.157	1044
Coal	850	-0.254	982
Coal	898	-0.311	1086
Gas	440	-0.233	430
Gas	398	-0.802	682

load fluctuation. In this case, the proportions of the three decomposition curves are shown in Fig. 4.

To evaluate the most suitable delivery method, AHP is utilized to determine the weight of each indicator to derive the comprehensive score. A three-level decision tree is proposed. The goal level represents the comprehensive indicators used to evaluate delivery methods. The criteria level includes eight secondary indicators, serving as the main basis for evaluation. The alternative level comprises the three contract delivery methods. A questionnaire is designed based on Saaty's nine-point scale and is distributed to professors, management experts, and power company experts active in the electricity market [79]. Based on their assessment of the current Chinese electricity market, experts score each indicator from 1 to 9, illustrating the importance of one indicator relative to another. Table 3 shows the detailed guide for experts to score according to Saaty's nine-point scale. The resulting 8x8 matrix is normalized to obtain the weight vector. To check the consistency of the experts' judgments, the consistency index (CI) and consistency ratio

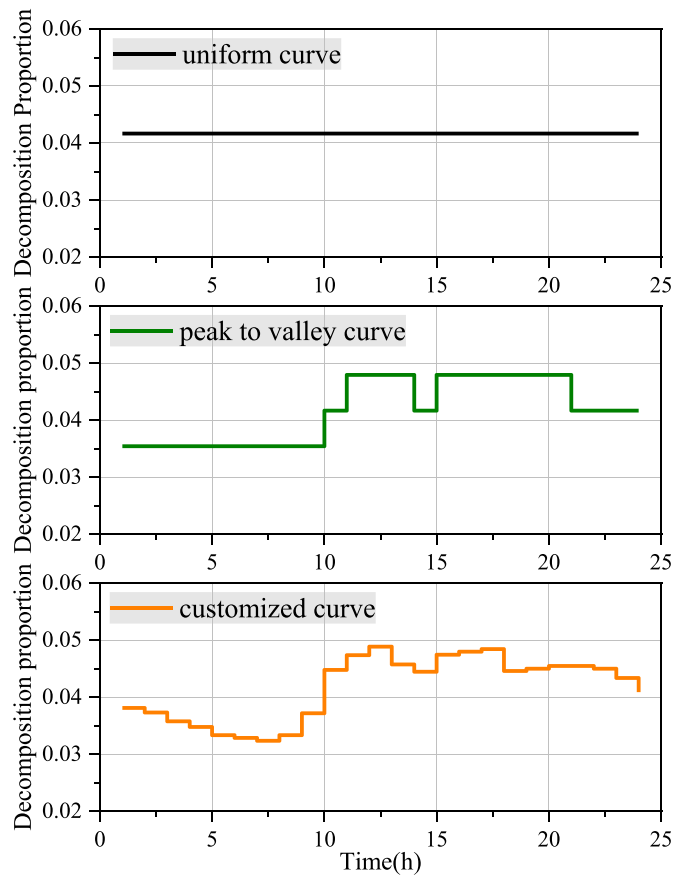


Fig. 4. Proportion of MLT decomposition curves.

(CR) are verified. The detailed calculation process of the AHP can be referenced from Ref. [80]. Table 4 shows the final weights assigned to each indicator.

5.2. Base scenario

In this section, the market equilibrium outcomes of the three delivery methods in the base scenario are analyzed. In the base scenario, GenCos participate in both the electricity markets and the carbon market. The carbon price is 100 CNY/ton. The MLT electricity accounts for 90 % of the total traded electricity, and the decomposition of the MLT contracts is customized decomposition.

Fig. 5 (a)–(c) illustrate the distribution of GenCos' bidding prices under various proportion of MLT contracts and different delivery methods. Fig. 5 (d) presents the variation of Nash equilibrium degrees during the simulation under the base scenario for physical delivery, financial delivery, and transitional delivery. As shown in Fig. 5 (a)–(c), the proportion of MLT significantly influences the bidding behavior of GenCos in the spot market. Specifically, GenCos with MLT contracts covering more than 90 % of their capacity tend to bid at lower prices in the spot market to secure sufficient generation for contract fulfillment. Conversely, GenCos with fewer MLT contracts are more likely to bid higher prices in the spot market to raise the clearing price, even if they fail to clear during certain periods, as their smaller contractual obligations are relatively easier to fulfill. This tendency is particularly pronounced in Fig. 5 (c), where transitional delivery imposes stricter penalties on unmet MLT contract obligations. The Nash equilibrium degree is employed to quantify the extent to which each agent deviates from the market equilibrium state, as defined in Ref. [81]. In this case study, the ideal Nash equilibrium degree is set at 40. As illustrated in Fig. 5 (d), the average deviation of the Nash equilibrium degrees for the three delivery methods is less than 3 % from the ideal value, demonstrating the model's effectiveness.

To further analyze the decision-making behavior, three GenCos from the three groups are randomly selected, as shown in Fig. 6. Fig. 6 (a)–(c) show the bid prices for units managed by GenCo 31 (Group 1), GenCo 1 (Group 2), and GenCo 19 (Group 3), respectively. Fig. 6 (d) shows the spot electricity market clearing prices under the three contract delivery methods. As can be seen from Fig. 6 (a)–(c), the bid prices of GenCos in the spot electricity market are clearly different between different groups. GenCo 31's cost is close to the system marginal cost and the strategy is to bid at the marginal price. GenCo 1 has a lower generation cost and the strategy is to bid a low price to guarantee more awarded power. Under financial delivery, GenCo 19 offers a high price during the valley hours (0:00–10:00) when the prices are low to avoid winning the bid, and offers a low price to get more awarded power when the prices are high. As GenCos under transitional delivery are required to fulfill their MLT contracts as much as possible or face penalties, the GenCos with contracts offer lower prices to ensure winning the bidding. Therefore, the average price of transitional delivery (451 CNY/MWh) is lower than that of physical delivery (674 CNY/MWh) and financial delivery (704 CNY/MWh).

Tables 5 and 6 show the market outcomes of three delivery cases. In the case of transitional delivery, some GenCos are fined for failing to fulfill their MLT contracts, which account for about 27 % of total revenues. As a result, the total revenues of GenCos are the lowest in the three delivery cases. Given that about 90 % of electricity is contracted through MLT agreements, the proportion of user-side spending in the spot market is only 10 %. Consequently, the total user-side spending is only subject to a 5 % change in different delivery methods. The carbon market can optimize the power generation structure in the electricity market, and

Table 3
The guide for filling out the questionnaire.

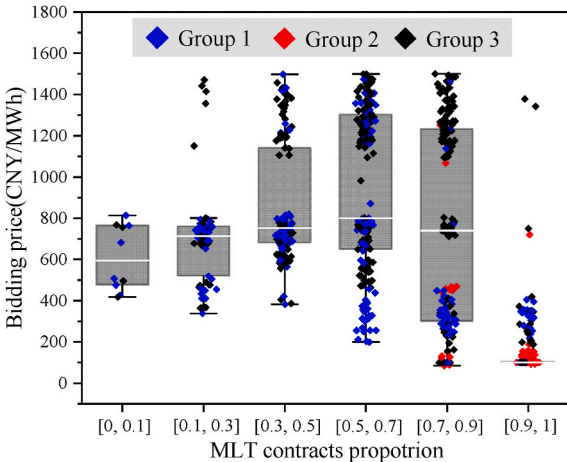
Preference	1	3	5	7	9	2,4,6,8
Definition	Equally preferred	Moderately preferred	Significantly preferred	Strongly preferred	Extremely preferred	Intermediate preferences

Table 4
Final weight of 8 indicators.

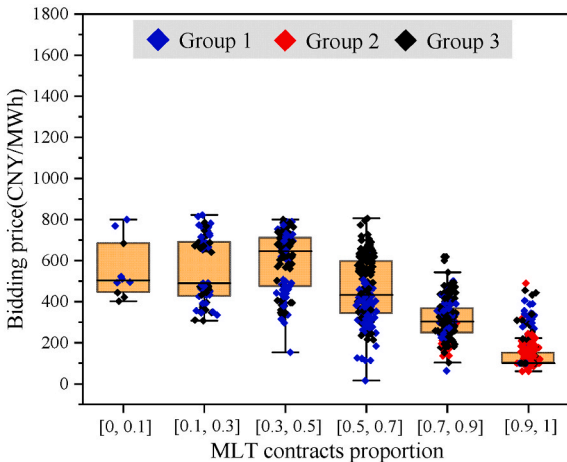
Indicator	Weight	Indicator	Weight
GenCo's revenues	0.153	Carbon emissions	0.150
Power purchase costs	0.127	MLT electricity correlation	0.041
Electricity prices	0.137	Carbon price conductivity	0.131
Market volatility	0.096	Fuel price conductivity	0.165

low-emission units like gas units can get more winning bids in the electricity market. However, in the case of physical delivery, the carbon market only has an impact on the revenues of GenCos from the carbon market, given that GenCos are obliged to generate electricity in strict accordance with their contracts. When the delivery constraints are

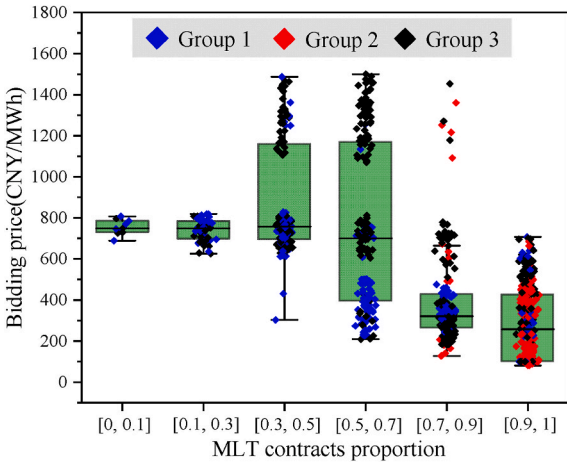
relaxed, GenCos bid in the spot market based on maximizing their revenues in both the electricity and carbon markets. In the case of transitional delivery, 27 GenCos have been identified as having failed to perform in accordance with MLT contracts. Of these, 90 % of gas-fired units (33 % of the total) are in violation, generating less electricity than the MLT contracts allow, thereby increasing carbon emissions. In the case of financial delivery, spot market electricity prices are higher, and gas-fired units win more bids, resulting in lower carbon emissions compared to transitional delivery and higher emissions compared to physical delivery.



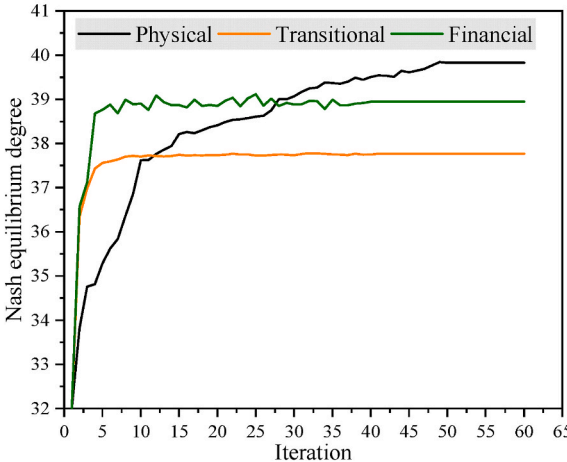
(a) Bidding prices of GenCos under physical delivery case



(b) Bidding prices of GenCos under transitional delivery case



(c) Bidding prices of GenCos under financial delivery case



(d) Nash equilibrium degrees under different delivery cases

Fig. 5. Bidding prices and Nash equilibrium degrees under 3 delivery methods.

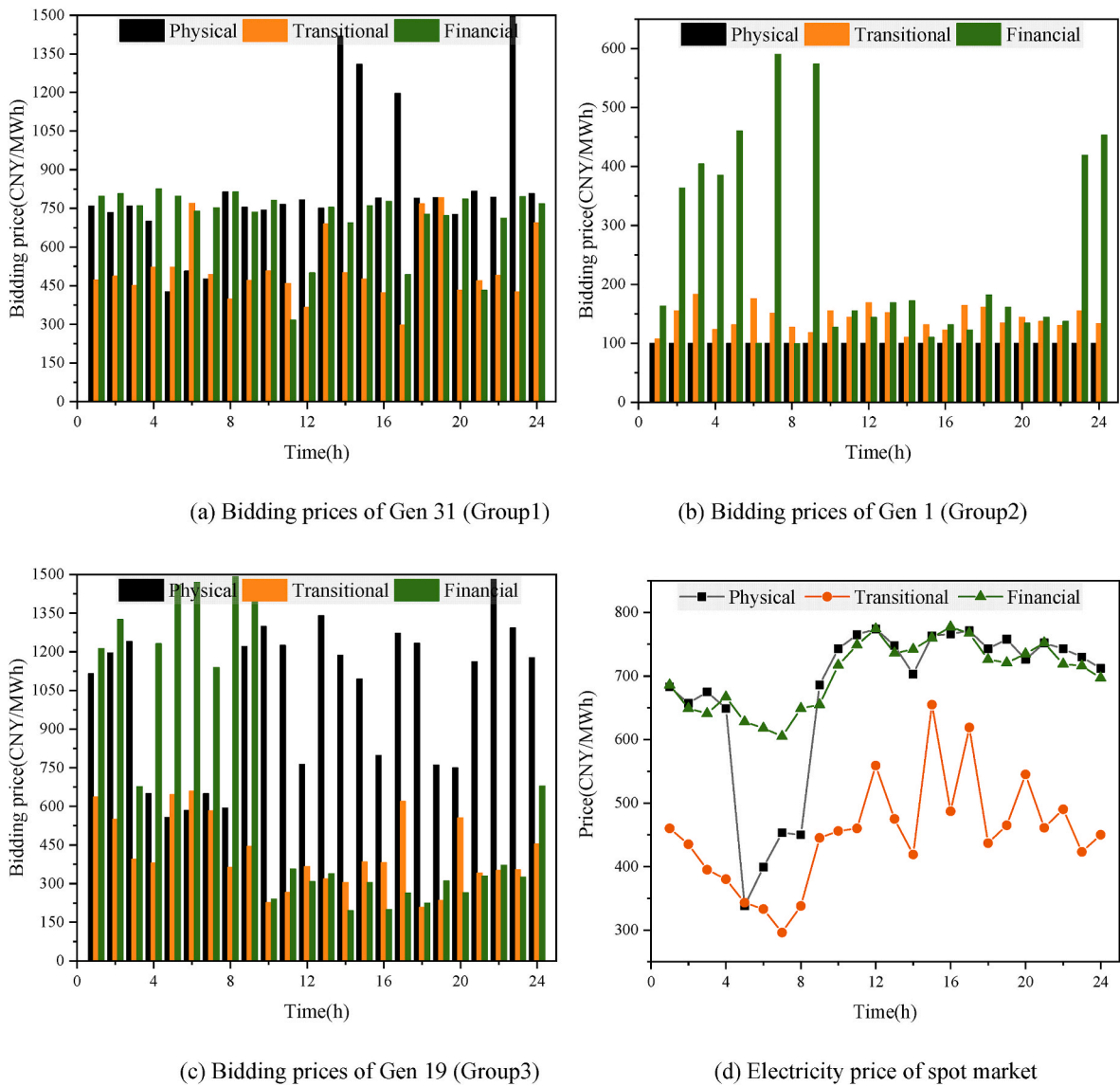


Fig. 6. Bidding prices and electricity prices under 3 delivery methods.

Table 5
Market outcomes under 3 delivery methods.

Delivery method	Total revenues (CNY)	Purchase costs (CNY)	Carbon emission (ton)
Physical	-19,443,626	464,934,420	713,733
Transitional	-56,752,672	444,321,624	718,752
Financial	-18,446,562	466,886,950	716,787

Table 6
Revenues of GenCos under 3 delivery methods.

Delivery method	MLT market (CNY)	Spot market (CNY)	Carbon market (CNY)	Fuel costs (CNY)	On/off costs (CNY)	Penalty (CNY)
Physical	402,799,379	62,135,040	-2,507,797	-481,870,248	0	0
Transitional	402,799,379	41,522,244	-2,811,747	-482,770,796	0	-15,491,754
Financial	402,799,379	64,087,570	-2,710,657	-482,622,855	0	0

5.3. Impacts of decomposition curve

Markets at different time scales are coupled with each other, and changes in MLT electricity market's outcomes such as prices and the composition of units may lead to changes in the spot electricity market and carbon market. In section 5.2, the initial indicators, including electricity prices, GenCos' revenues, power purchase costs, and carbon emissions, have been analyzed. This section proposes a "market volatility" indicator *Vol*, which represents the magnitude of changes in spot

market outcomes, as shown below:

$$Vol = \omega_1 pr + \omega_2 re + \omega_3 ue + \omega_4 em \quad (83)$$

$$pr = \sum_{t=1}^T \left(\sqrt{\frac{1}{n^s} \sum_{s=1}^{n^s} (\lambda_{s,t}^D - \bar{\lambda}_{s,t}^D)^2} / \lambda_{Base,t}^D \right) / T \quad (84)$$

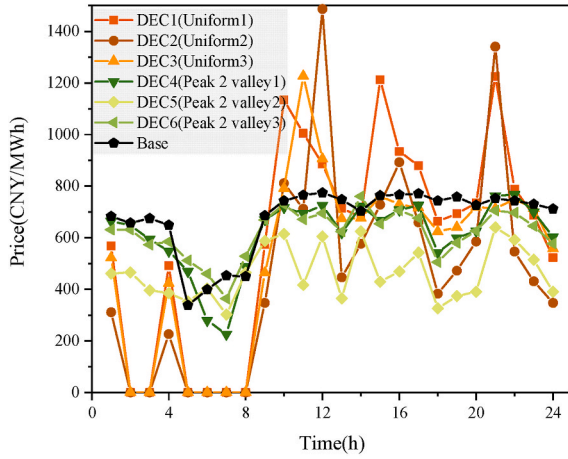
$$re = \sqrt{\frac{1}{n^s} \sum_{s=1}^{n^s} \left(\sum_{i=1}^N f_{s,i,t}^D - \sum_{i=1}^N \bar{f}_{s,i,t}^D \right)^2} / \sum_{i=1}^N f_{Base,i,t}^D \quad (85)$$

$$em = \frac{\sqrt{\frac{1}{n^s} \sum_{s=1}^{n^s} \left(\sum_{i=1}^N \sum_{t=1}^T (\delta_i P_{s,i,t}^{D^2} + \gamma_i P_{s,i,t}^D) - \sum_{i=1}^N \sum_{t=1}^T (\bar{\delta}_i P_{s,i,t}^{D^2} + \bar{\gamma}_i P_{s,i,t}^D) \right)^2}}{\sum_{i=1}^N \sum_{t=1}^T (\delta_i P_{s,i,t}^{D^2} + \gamma_i P_{s,i,t}^D)} \quad (86)$$

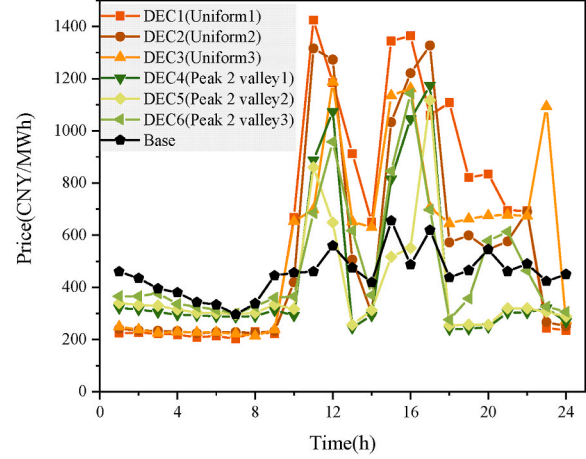
Where, pr represents price volatility, re represents the volatility of GenCos' revenues, ue represents the volatility of power purchase costs, and em represents carbon emission volatility. $\omega_1, \omega_2, \omega_3, \omega_4$ represent the weight of each sub-indicator, s represents the scenario index, and n^s represents the number of scenarios.

DEC1-3: Decomposition curve is uniform. In DEC1, Group1 has the fewest contracts. In DEC2, Group2 has the fewest contracts. In DEC3,

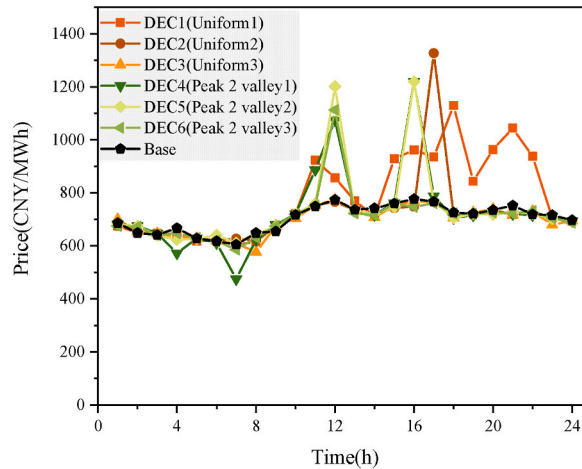
$$ue = \frac{\sqrt{\frac{1}{n^s} \sum_{s=1}^{n^s} \left(\sum_{i=1}^N \sum_{t=1}^T (P_{s,i,t}^L O_{i,t}^L + P_{s,i,t}^D \lambda_{s,t}^D) - \sum_{i=1}^N \sum_{t=1}^T (\bar{P}_{s,i,t}^L O_{i,t}^L + \bar{P}_{s,i,t}^D \bar{\lambda}_{s,t}^D) \right)^2}}{\sum_{i=1}^N \sum_{t=1}^T (P_{s,i,t}^L O_{i,t}^L + P_{s,i,t}^D \lambda_{Base,t}^D)} \quad (86)$$



(a) Electricity prices in physical delivery



(b) Electricity prices in transitional delivery



(c) Electricity prices in financial delivery

Fig. 7. Electricity prices in different scenarios.

Table 7

Electricity price volatility under 3 delivery methods.

Index	Physical delivery	Transitional delivery	Financial delivery
Standard deviation (CNY/MWh)	168.13	157.82	55.99
Price volatility (%)	27.00	33.41	7.58

Group3 has the fewest contracts.

DEC4-6: Decomposition curve is peak to valley. In DEC4, Group1 has the fewest contracts. In DEC5, Group2 has the fewest contracts. In DEC6, Group3 has the fewest contracts.

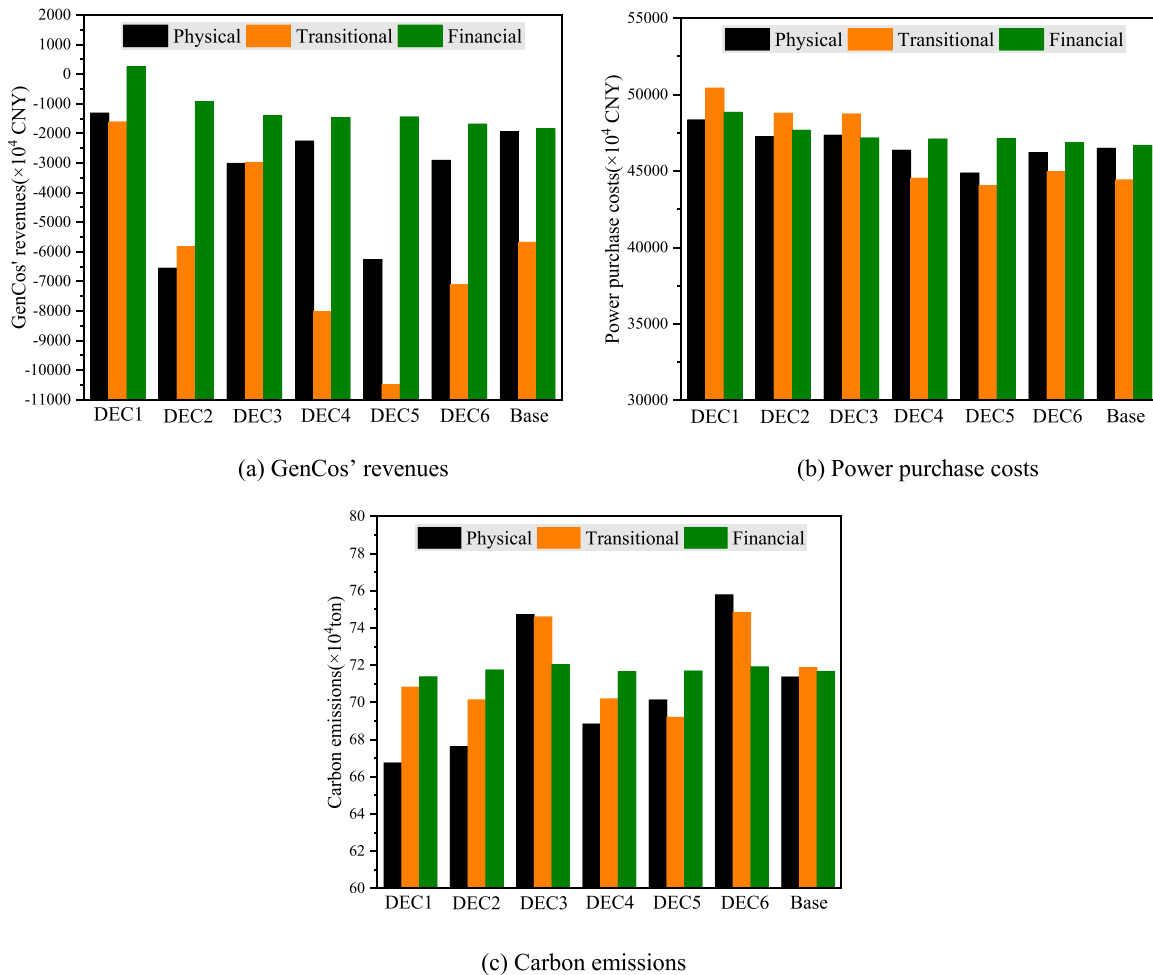
Fig. 7 shows the electricity prices in spot market for different scenarios under physical delivery, transitional delivery and financial delivery, respectively. From Fig. 7 (a), it can be seen that in the DEF1-3 scenarios, the MLT contracts are uniformly decomposed, and the prices fluctuate more under physical delivery, from 0 CNY/MWh (2:00–3:00, 5:00–8:00) to 1500 CNY/MWh (12:00). In contrast, the DEF4-6 scenarios, where the MLT contracts are decomposed according

Table 9

Market volatility under 3 delivery methods.

Index	Physical delivery	Transitional delivery	Financial delivery
Vol (%)	33.31	22.71	11.23

to the peak to valley curve, have a similar trend of prices as the Base scenario, with an overall smoother (302–760 CNY/MWh). From Fig. 7 (a), it can be seen that under transition delivery methods, the average prices of DEF1-3 scenarios are lower than that of DEF4-6 during valley hours (1:00–10:00), but higher than those of DEF4-6 during non-valley hours (11:00–24:00). Therefore, the same conclusion as under physical delivery, the prices are more volatile under the DEF1-3 decomposition approach. However, there is no significant change in the prices in the DEC1-6 scenarios under financial delivery, which are overall smoother, as shown in Fig. 7 (c). Table 7 shows the price volatility under the 3 delivery methods. The 24-h average standard deviation and volatility under financial delivery are the smallest, and the average volatility of each hour is only 7.58 %, which indicates that the electricity prices

**Fig. 8.** Market clearing outcomes in different scenarios.**Table 8**

Volatility of results under 3 delivery methods.

Volatility	Physical delivery	Transitional delivery	Financial delivery
GenCos' revenues(%)	99.59	48.91	35.64
Power purchase costs(%)	2.18	5.53	1.44
Carbon emissions(%)	4.47	2.99	0.27

under financial delivery have better stability.

Fig. 8 (a)–(c) show the GenCos' revenues, power purchase costs, and carbon emissions in different scenarios of the 3 delivery cases. As can be seen from Fig. 8 (a)–(c), the GenCos' revenues are lowest in the case of transitional delivery in most scenarios. The revenue gap on the GenCos in different delivery cases in the DEC3 is the smallest at 1615.28×10^4 CNY, while the revenue gap on the GenCos in the DEC4 is the largest at 9043.04×10^4 CNY. Since most of the power purchase costs (about 90 %) are fixed by MLT contracts, the total cost of power purchased by the user remains largely unchanged in different scenarios and delivery cases. During the transition from a physical to a financial delivery, the change in carbon emissions across different scenarios exhibits a consistent pattern, which is referred to as the reverse adjustment property in this paper. Carbon emissions gradually increase from physical to financial delivery when more low-emission units are contracted, and conversely, carbon emissions gradually decrease from physical to financial delivery when fewer low-emission units are contracted. This is the result of the interaction between a regulated market and an open market. Fig. 8 also shows that the results of financial delivery are more stable. Table 8 shows the volatility in GenCos' revenues, power purchase costs, and carbon emissions in different scenarios and delivery cases. The decomposition curve and composition of MLT contracts have a significant impact on the revenues of GenCos. The maximum volatility is 99.59 % in physical delivery case, while the minimum volatility is 35.64 % in financial delivery case. In contrast, the volatility of power purchase cost and carbon emissions is relatively insignificant compared to the

volatility of GenCos' revenues, with a maximum of only 5.53 %.

In summary, Table 9 shows the market volatility indicators under the 3 policy delivery methods.

5.4. Impacts of medium and long-term contract percentage

Not only do changes in the decomposition curves and composition of MLT contracts cause changes in electricity market outcomes, but the amount of MLT contracts also affects electricity market outcomes. Based on the initial indicators analyzed in Section 5.2, this article analyzes the sensitivity of the electricity market to MLT traded electricity in different policy delivery cases, and proposes a derived indicator for MLT electricity correlation X^{ML} , as shown below:

$$X^{ML} = X^{pr} + X^{re} + X^{ue} + X^{em} \quad (88)$$

$$X^{pr} = \Delta \left(\frac{\left(\frac{\sum_{t=1}^T \lambda_{s,t}^D}{T} \right) / T - \max \left(\left(\frac{\sum_{t=1}^T \lambda_{s,t}^D}{T} \right) / T \right)}{\max \left(\left(\frac{\sum_{t=1}^T \lambda_{s,t}^D}{T} \right) / T \right) - \min \left(\left(\frac{\sum_{t=1}^T \lambda_{s,t}^D}{T} \right) / T \right)} \right) / \Delta \left(\frac{P_{s,i,t}^L}{P_{s,i,t}^D} \right) \quad (89)$$

$$X^{re} = \Delta \left(\frac{\sum_{i=1}^N f_{s,i,t}^D - \max \left(\sum_{i=1}^N f_{s,i,t}^D \right)}{\max \left(\sum_{i=1}^N f_{s,i,t}^D \right) - \min \left(\sum_{i=1}^N f_{s,i,t}^D \right)} \right) / \Delta \left(\frac{P_{s,i,t}^L}{P_{s,i,t}^D} \right) \quad (90)$$

$$X^{ue} = \Delta \left(\frac{\sum_{i=1}^N \sum_{t=1}^T \left(P_{s,i,t}^L O_{i,t}^L + P_{s,i,t}^D \lambda_{s,t}^D \right) - \max \left(\sum_{i=1}^N \sum_{t=1}^T \left(P_{s,i,t}^L O_{i,t}^L + P_{s,i,t}^D \lambda_{s,t}^D \right) \right)}{\max \left(\sum_{i=1}^N \sum_{t=1}^T \left(P_{s,i,t}^L O_{i,t}^L + P_{s,i,t}^D \lambda_{s,t}^D \right) \right) - \min \left(\sum_{i=1}^N \sum_{t=1}^T \left(P_{s,i,t}^L O_{i,t}^L + P_{s,i,t}^D \lambda_{s,t}^D \right) \right)} \right) / \Delta \left(\frac{P_{s,i,t}^L}{P_{s,i,t}^D} \right) \quad (91)$$

$$X^{re} = \Delta \left(\frac{\sum_{i=1}^N \sum_{t=1}^T \left(\delta_i P_{s,i,t}^{D^2} + \gamma_i P_{s,i,t}^D \right) - \max \left(\sum_{i=1}^N \sum_{t=1}^T \left(\delta_i P_{s,i,t}^{D^2} + \gamma_i P_{s,i,t}^D \right) \right)}{\max \left(\sum_{i=1}^N \sum_{t=1}^T \left(\delta_i P_{s,i,t}^{D^2} + \gamma_i P_{s,i,t}^D \right) \right) - \min \left(\sum_{i=1}^N \sum_{t=1}^T \left(\delta_i P_{s,i,t}^{D^2} + \gamma_i P_{s,i,t}^D \right) \right)} \right) / \Delta \left(\frac{P_{s,i,t}^L}{P_{s,i,t}^D} \right) \quad (92)$$

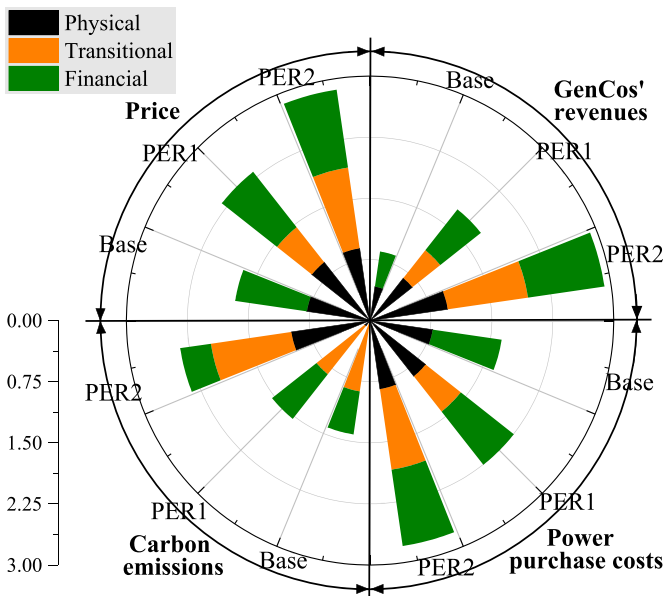


Fig. 9. Market outcomes under 3 delivery methods in different scenarios.

Where, X^{pr} represents the magnitude of the change in electricity prices with the proportion of MLT electricity, X^{re} represents the magnitude of the change in GenCos' revenues, X^{ue} represents the magnitude of the change in power purchase costs, and X^{em} represents the magnitude of the change in carbon emissions.

PER1: MLT electricity accounts for 80 % of total electricity consumption.

PER2: MLT electricity accounts for 70 % of total electricity consumption.

Fig. 9 shows the normalized results of GenCos' revenues, power purchase costs, carbon emissions, and average electricity prices in different PER scenarios. The market equilibrium outcomes of physical delivery and transitional delivery will eventually be consistent with those under financial delivery as the ratio gradually decreases. This is due to the fact that in physical delivery case, the MLT contract power

Table 10
MLT electricity correlations under 3 delivery cases.

Index	Physical delivery	Transitional delivery	Financial delivery
X^{pr} (%)	2.72	5.00	2.58
X^{re} (%)	0.54	5.00	0.26
X^{ue} (%)	4.85	0.53	−0.67
X^{em} (%)	0.55	4.99	0.37
X^{ML} (%)	8.65	15.52	2.54

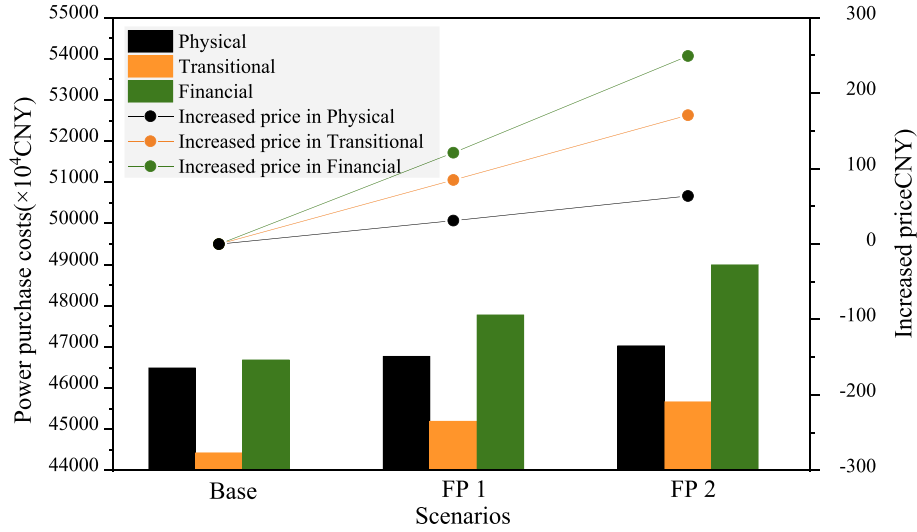


Fig. 10. Fuel price conductivity and increased power purchase costs in different scenarios.

constraint undergoes a change from a tight constraint to a loose constraint when the amount of enforced power is less than the amount of power cleared in the spot market under unconstrained enforcement. In transitional delivery case, the policy penalty constraint on MLT contracts is one-way, and GenCos are not penalized for generating more electricity in the spot electricity market than they would have under MLT contracts. After the proportion of electricity traded in the MLT electricity market falls below 70 %, there is little difference between the different delivery cases. Therefore, the physical and transitional delivery methods currently implemented in China allows for a gradual transition to financial delivery by reducing the ratio requirement. Table 10 shows the MLT electricity correlations X^{ML} in the 3 delivery cases, with the highest MLT correlation at 15.52 %⁻¹ in transition delivery cases, indicating that the electricity market is most affected by the change in the proportion of MLT transactions from 90 % to 70 %. On the contrary, the electricity market shows better stability in the case of financial delivery.

5.5. Impacts of fuel market

In this section, the relationship between the electricity market and the fuel market in the three delivery cases is analyzed. When fuel price

risks, the conductivity of fuel price differs across various delivery cases, exhibiting a gradual incremental rise in fuel price conductivity from physical to financial delivery. Therefore, the derived indicator which is defined as fuel price conductivity in this paper is given in the following equation:

$$\rho^{fuel} = \frac{\Delta \lambda_{s,t}^D}{\Delta c_s^{fuel}} \times 100\% \quad (93)$$

Where, c_s^{fuel} represents the fuel price in scenario s.

FP1: Fuel price rises to 1000 CNY/ton.

FP2: Fuel price rises to 1200 CNY/ton.

Fig. 10 shows fuel price conductivity in three delivery cases. In this paper, it is assumed that the gas price of a gas-fired generator is converted to the coal price, which fluctuates in parallel with the fuel cost of a coal-fired generator. Additionally, the electricity price is considered as a 24-h weighted average price. When fuel prices rise, spot electricity market clearing prices increase in all three delivery cases. However, the degree of increase varies with the delivery, with the conductivity rising in the order of 15.93 %, 42.71 %, and 62.28 % from physical to financial delivery respectively. The transmission of fuel prices to electricity prices exhibits a sequential increase in conductivity during the transition from

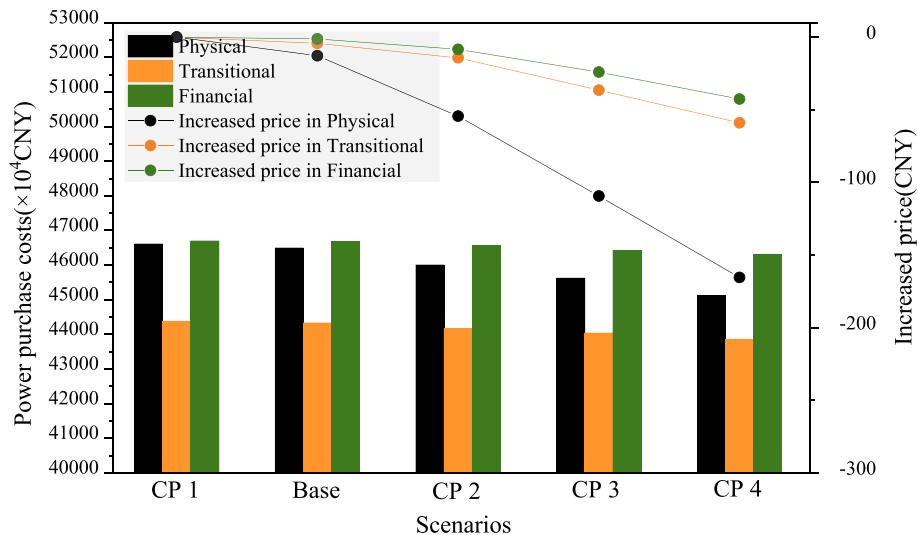


Fig. 11. Carbon price conductivity in different scenarios under dynamic CEI.

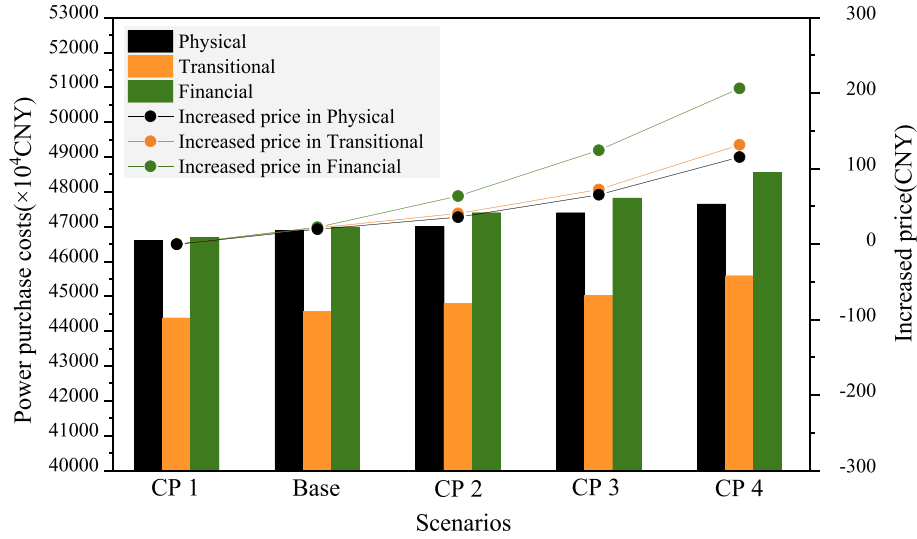


Fig. 12. Carbon price conductivity in different scenarios under static CEI.

physical delivery, through transitional delivery, to financial delivery. This sequential pattern suggests that the policies associated with physical and transitional delivery partially hinder the transmission of fuel price fluctuations to the user side.

5.6. Impacts of carbon market

Electricity market prices are equally affected by carbon prices in addition to fuel prices. The carbon price conductivity ρ^{carbon} is defined as shown below:

$$\rho^{carbon} = \frac{\Delta \lambda_{s,t}^D}{\Delta \lambda_s^{carbon}} \times 100\% \quad (94)$$

Where, λ_s^{carbon} represents the carbon price in scenario s.

CP1: Carbon market is not considered.

CP2: Carbon price rises to 300 CNY/ton.

CP3: Carbon price rises to 600 CNY/ton.

CP4: Carbon price rises to 1000 CNY/ton.

As shown in Fig. 11, when the carbon prices increase gradually, carbon emissions decrease. However, the price of electricity undergoes a decrease, with a decreasing rate of decrease observed from physical to financial delivery. When the carbon price is at 0 CNY/ton, indicating the absence of a carbon market, the cost of purchasing electricity increases compared to the reference scenario (coal price = 800 CNY/ton, gas price = 2.2 CNY/m³, carbon price = 100 CNY/ton). The price of electricity exhibits a gradual decline as the carbon price increases. Notably, this reduction is most significant in the physical delivery, followed by the transitional delivery, and the least significant in the financial delivery. The negative conductivity of carbon prices from physical to financial delivery is -16.4 %, -5.3 %, and -3.3 %, respectively, which is contrary to the common theory of carbon price conductivity [31].

This is because, on the one hand, all the allowances in Chinese national carbon market are issued free of charge, and the burden of

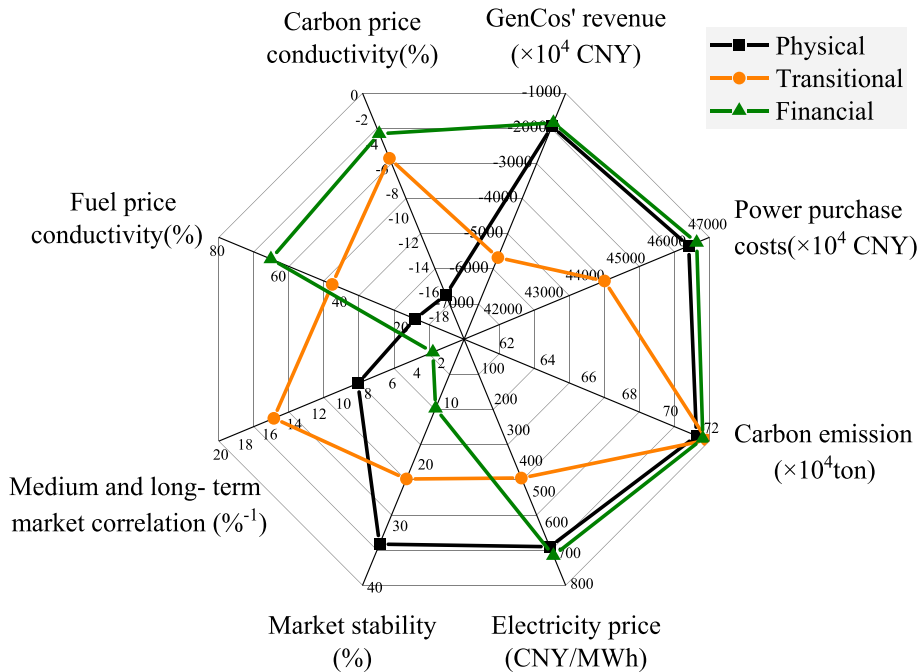


Fig. 13. Market performance evaluation matrix of the Guangdong case.

emission reduction on GenCos is not heavy. GenCos are not all buyers of allowances, but GenCos with low carbon emission intensity are sellers of allowances in the carbon market. When the carbon price = 100 CNY/ton, coal-fired GenCos with the number {3,4,7,10,13,15} and all gas-fired GenCos are sellers of allowances in the carbon market. On the other hand, under the dynamic CEI characteristic, the higher the load factor of generators, the lower the emission intensity, and the role of some GenCos changes from allowance buyers to allowance sellers, which can gain from the carbon market. For example, as the carbon prices increase from 100 CNY/ton to 1000 CNY/ton, coal-fired GenCos numbered {8,14} switch to allowance sellers.

To guarantee more winning bids, GenCos declare lower prices. In the case of physical delivery, it is imperative that GenCos do not produce less power in the spot electricity market than in the MLT contracts, and that the load factor is higher when the units' strategies are consistent. This is to ensure that the GenCos win more bids and obtain more CEAs, which in turn will lower the offer price. In contrast, in the case of financial delivery, GenCos do not strictly implement the MLT contracts. Some of GenCos' strategies are to ensure a high load factor to get more revenue from the carbon market. Conversely, some GenCos, such as marginal units, are unable to ensure a high load factor strategy to get more revenue from the electricity market, and still declare high prices in the electricity market. Consequently, the negative conductivity of carbon price from the physical delivery to the financial delivery is decreasing gradually.

Fig. 12 shows the impact of the carbon market price on the electricity market at static CEI in different CP scenarios of 3 delivery cases. In contrast to the findings observed under dynamic CEI, the electricity price under static CEI increases with the carbon price from physical to financial delivery. This increase corresponds to carbon price conductivity of 13.5 %, 14.8 %, and 21.2 %, respectively. The reason for this phenomenon under static CEI is that surplus CEAs do not scale with output. Consequently, the cost of generating electricity increases for generators with emission intensities higher than the standard as the carbon price rises, leading to an increase in their offer prices.

5.7. Impacts of policy execution

The analysis of the indicator system in sections 5.2-5.6 shows that different aspects of the electricity market are affected by the transition in the way contracts are implemented from physical to financial. Fig. 13 shows the indicators analyzed in section 5.2 and the derived index analyzed in Sections 5.3-5.6. **From the GenCo's perspective**, the total revenues of GenCos are the largest under financial delivery, while the fuel price conductivity is the highest at 63 %, which can transfer more

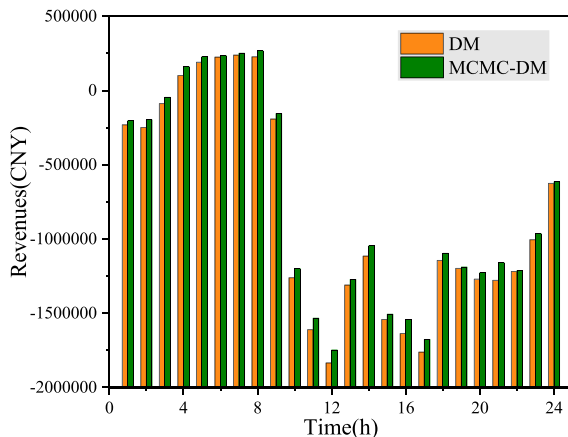
than half of the price fluctuation risk to the user side when the fuel price rises, meanwhile, the market volatility and correlation are the smallest under financial delivery, 11.13 %, and 2.54 % respectively, indicating that the spot market has high stability, which is favorable for the enterprise risk management. **From the user's perspective**, electricity prices are the lowest in transitional delivery case and total power purchase costs are 4.83 % lower compared to financial delivery, but physical delivery is more conducive to user participation in the electricity market. On the one hand, the spot market is less affected by the outcome of MLT transactions, on the other hand, fuel price fluctuations are barely transmitted to users (about 16 %) and the electricity prices fall the most when the carbon price rises. **From the government's perspective**, the transitional delivery currently being implemented in China is a compromise between the effects of physical delivery and financial delivery. In the process of physical delivery to financial delivery, the risk of market price fluctuations is gradually shifted from the power generation side to the user side, but the electricity market's overall ability to resist the risk is gradually rising, and the market price is more stable.

To obtain a comparison of the three delivery methods, the results shown in Fig. 13 are normalized. Subsequently, the normalized indicators are multiplied by their respective weights to obtain the weighted scores of each indicator. Notably, metrics such as power purchase costs, carbon emissions, and electricity prices are preferable when lower. To align these metrics with the higher-is-better criterion, they are multiplied by negative one. Finally, the weighted scores of all indicators are aggregated to derive the composite score. The results indicate that transitional delivery achieves the highest composite index at 19.7 %, followed by financial delivery at 14.3 %. Physical delivery shows the lowest composite index at 11.7 %.

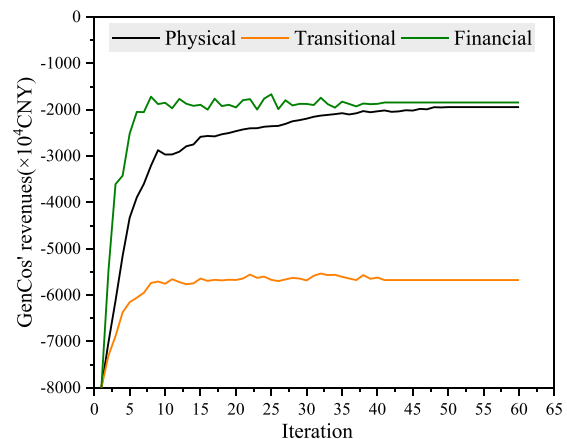
The optimization results and Convergence curves of MCMC-DM algorithm are shown in Fig. 14. More revenues are obtained using the MCMC-DM than using the DM. The MCMC-DM is found to be better at finding globally optimal solutions.

As shown in Figs. 15 and 16, the case of Jiangsu electricity market is used to validate the effectiveness and robustness of the model in other provincial electricity markets. The Jiangsu electricity market case is designed based on the installed capacity distribution in Jiangsu Province [82], with generator parameters randomly selected from Tables 1 and 2. The MLT price is set at 512 CNY/MWh, which is higher than the MLT price of 496 CNY/MWh in the Guangdong electricity market. Fig. 15 illustrates the curve results of MLT power decomposed into the spot market after market competition.

Fig. 16 presents the market performance evaluation matrix for the Jiangsu electricity market case. Compared to Guangdong, Jiangsu case features a higher proportion of gas-fired units participating in the



(a) Optimization results of MCMC-DM



(b) Convergence curves of MCMC-DM

Fig. 14. Optimization results and convergence curves of MCMC-DM.

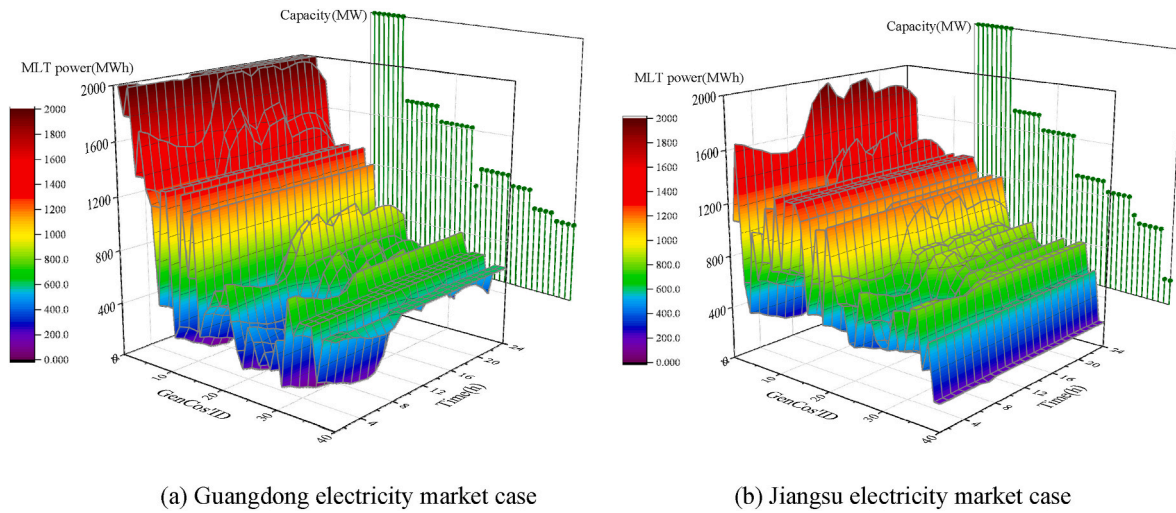


Fig. 15. MLT contracts and capacities of GenCos.

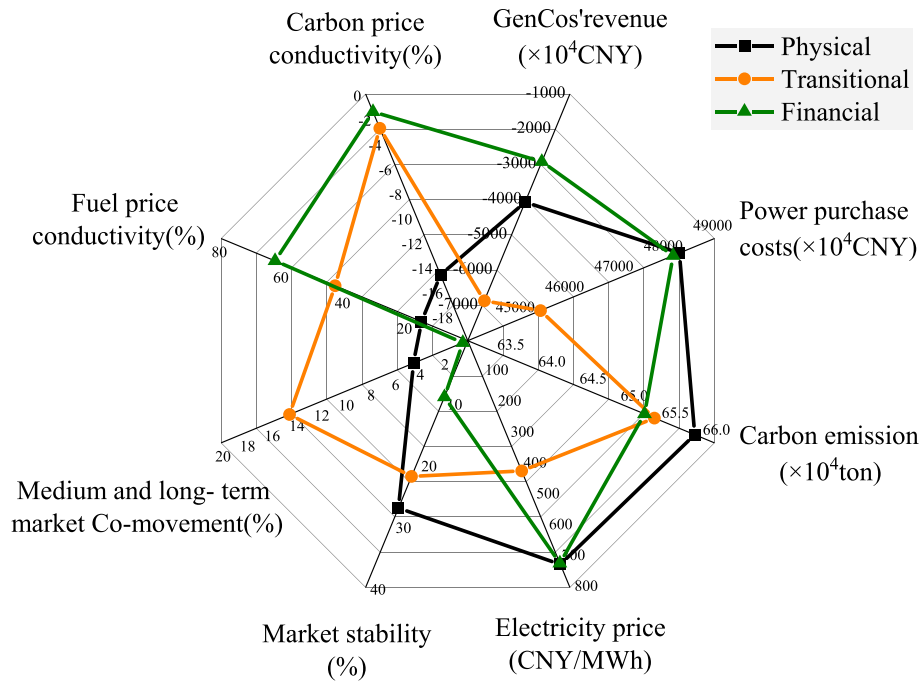


Fig. 16. Market performance evaluation matrix of the Jiangsu case.

market, resulting in higher electricity prices and increased power purchase costs for users. While the results for Jiangsu align with those for Guangdong in terms of the differences among the three delivery methods, the higher MLT market price in Jiangsu reflects more pronounced conservative strategies of GenCos under transitional delivery, with the per MWh power price being 28.8 CNY lower than that in Guangdong. The results of the composite index for the Jiangsu case indicate that transitional delivery achieves the highest value at 29.5 %, followed by financial delivery at 19.2 %, while physical delivery records the lowest value at −19.8 %. The results from the Jiangsu case further validate the robustness and applicability of the proposed model. Despite regional differences in generation mix and MLT pricing, the transitional delivery method continues to achieve a balanced performance in terms of market efficiency and risk transmission.

6. Conclusions

During the transition to a market-based and decarbonized power system, the delivery methods of MLT contracts has become a topic of increasing debate. With the growing share of renewable energy and the evolving generation structure of the power system, selecting the optimal contract delivery method based on the system's development stage is crucial for achieving carbon neutrality goals. This paper proposes a bi-level game-based simulation model to analyze the transitional process in electricity and carbon markets. Then, the EPEC model formed by multiple interacting bi-level game models is solved by using MCMC-DM algorithm. A case study of Guangdong electricity market validates the model's effectiveness. The results of the simulation show that:

Firstly, the transitional delivery method for MLT contracts is most suitable for the current stage of Chinese power system. In the transitional delivery scenario, the comprehensive indicator outperforms that of the physical delivery scenario by 8 % and the financial delivery

scenario by 5.4 %. This suggests that the transitional method strikes a balance between market efficiency and risk management, making it an optimal choice for the current transitional phase of the electricity market. It is recommended that policymakers continue supporting the transitional delivery method, while gradually moving towards more market-driven mechanisms to enhance efficiency without overwhelming market participants with risks.

Secondly, GenCos' revenues and power purchase costs are minimized under transitional delivery. Electricity market volatility is greatest under physical delivery, with GenCos almost entirely burdened with the risk of fuel price and carbon price fluctuations. On the contrary, electricity market volatility is minimized under financial delivery, and most of the risk of fuel price and carbon price fluctuations is passed on to users. Therefore, it is imperative for the Government to adjust the delivery method in the light of the revenues and the risk-bearing capacity of the GenCos and the users. It is suggested that policymakers consider the risk-bearing capacity of both GenCos and consumers when adjusting the delivery method. A gradual transition to financial delivery could create a more stable market environment while reducing GenCos' exposure to volatile fuel and carbon prices.

Thirdly, considering dynamic CEI, the electricity prices decrease as the carbon prices increase; by contrast, the electricity prices increase as the carbon prices increase considering static CEI. Failure to account for these factors could lead to inefficient pricing and distortions in the market. Therefore, it is crucial for market participants to incorporate dynamic CEI into their decision-making process to ensure optimal market outcomes.

In future research, models for both the electricity and carbon markets should incorporate risk factors. Methods such as Information Gap Decision Theory (IGDT) and risk-based stochastic approaches can be employed to address uncertainties [83,84], including those related to renewable energy generation, load fluctuations [85], and the behavior of non-electricity sectors in the carbon market.

CRedit authorship contribution statement

Peng Wang: Writing – original draft, Supervision, Resources, Conceptualization. **Wentao Wang:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Kai Jiang:** Writing – review & editing, Visualization, Validation, Supervision, Project administration. **Yixin Cheng:** Writing – review & editing, Investigation, Formal analysis. **Tengxi Zhang:** Visualization, Software, Resources, Methodology. **Xuesong Li:** Supervision, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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